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Chapter One

The Nature and Functions of money

1. Introduction

Most of the monetary theories are written in inexplicit language, which not only restricts this discipline to the so called Politico- Economists of the country, but also keeps it hidden from the common masses. The inability to understand the complexity of theories has misleadingly led many to believe that money originally started as a government initiative rather than a market phenomenon. It was the —forces of market, its discipline and needs that led to the invention of money, and eventually the evolution also kicked in by the same forces. However this auto evolution came to a halt when governments actively started interfering with respect to what money actually should be. The role of authorities itself changed from —the regulator of the money approved by the markets to —the creator and issuer of the money coerced upon the markets.

1.1. Meaning and functions of money

The importance of money for modern economies is known by the functions that it performs. As a matter of fact, it is difficult to think of an economy these days which may work without the use of money.

Therefore, money is put to different uses in economic society. It has already been indicated that the fundamental purpose of money is to serve as a medium of exchange and as a unit of account or what is generally called as a measure of value. These are its fundamental or primary functions. In the secondary capacity, money serves as a standard of deferred payments and as a store of value. It also performs several other functions in a modern economy.

In general, Kinley has classified the functions of money into three groups:

- I. **Essential or Primary**
- II. **Derived (Secondary) or Subsidiary; and**
- III. **Contingent**

I. Primary Functions of Money

Primary or essential functions of money are also known as the original, the fundamental or the basic functions of money. These are the functions which have been performed by money in all

countries at all times and under all conditions. Therefore, the medium of exchange' and the 'common measure of value' functions of money are included in this category.

A. Money as a medium of Exchange: - Although money has no power to satisfy human wants directly, it commands power to purchase those things which have utility and satisfy human wants. This gives money a unique place in a society. The use of money as a medium of exchange promotes economic efficiency by illuminating much of the time spent in exchange of goods and services.

- Money has given considerable freedom of action to the buyers and the sellers of goods with regard to the quantities and qualities of the goods they desire to buy and sell; the time and place of transactions and the parties with whom they choose to deal.
- Money functioning as a medium of exchange has, therefore, perfected the price mechanism. It has also greatly assisted the productive operations. Each factor of production in a capitalistic society receives an amount of money income in return for his contribution to production. The share of each factor of production in a capitalistic society receives an amount of money income in return for his contribution to production. The share of each factor of production in the output of industry is immediately determined, and it has a wide choice as to its disposal.
- On the other hand, money decomposes the single transaction of barter into two separate transactions of sale & purchase and illuminates the need for double coincidence of wants. The decomposition is both in time and space. These allow for the use of division of labor which boaster productivity in a society. It also allows for the freedom of choice.
- However, the acceptance of money as a medium of exchange is a matter of social convention because money is created by the society (here money is the agreement by the society). So each person accepts money as a means of payment because he is confident that others will accept it in payment for him. The social convention could either be established through legal or other means. Therefore, a commodity to be accepted as money in order to serve as a medium of exchange must meet at least the following criterion:
 - It must be easily standardized.
 - It must be widely accepted
 - It must be divisible so that it is easy to make a change.
 - It must be easy to carry

- It must not deteriorate quickly (be durable)

B. Money as a measure value

Money serves as the unit in terms of which the value of all goods and services is measured and expressed as a price since every commodity is exchanged against money, the values of all commodities and services are expressed only in terms of money.

The pricing mechanism, which helps us in obtaining efficient allocation of resources in the economic system, is dependent on the measure of value function of money. Without the price system, the values of goods and services would have to be expressed in terms of all other available goods and services.

- All business organizations would be severely handicapped were a monetary unit not available as a unit of account. Assets and liabilities of all kinds, incomes and expenditures of all kinds, all kinds of economic values, profits and losses and all kinds of remunerations paid to the factors of production are stated in terms of common monetary units.

Here, we may say that money measures the value, just as we say inches and meters are used to measure linear distance. All units of physical measurement are themselves constant quantities, while money is like a yardstick whose length vary from time to time. Variations in the length of money yardstick can be seen in the changing price levels and changing purchasing power of money. Any increase or decrease in the quantity of money in circulation is likely to affect its purchasing power; which is to say that the value of goods and services in terms of money is likely to vary. All this mean that money serves quite satisfactorily as a standard of value at a given time and within a limited area.

In general money enables and orderly pricing system which is essential for □

Rational economic calculation & choice.

- Transmitting economic information between individuals.

II. Subsidiary Functions of Money

In addition to its two primary functions, money performs the secondary functions. These secondary functions of money are, discussed as follows:

1. Money as standard of deferred payments

Money comes into general use as a medium of payments and as a unit of account not only for **immediate transactions**; it also serves as a standard of deferred payments when obligations to make future payments are stated in terms of it. These obligations have their origin in two general types of transactions:

- The first is that in which one contracting party agrees to deliver fixed amounts of goods, services or securities at some future time in exchange for an agreed upon sum of money to be paid in the future.
- The second type of transactions giving rise to the use of money as a standard of deferred payments includes **credit or debit transactions**, in which the creditor parts with things of value at one time in return for which the debtor promises to repay money at some future date.

Regardless of what is bought on account or what is borrowed, these debts are almost invariable stated in terms of money.

Money has successfully performed the function of a standard of deferred payments because:

- ☐ Its value is relatively more stable than that of other commodities.
- ☐ The element of durability is higher as compared to other commodities; and
- ☐ It possesses the quality of general acceptability.

In the absence of money, lending would be difficult, rather impossible. It may, however, be mentioned in this connection, that **as a standard of deferred payments also money suffers from certain shortcomings and the main drawback is that its value is not wholly stable**. Obviously, when the value of money itself depreciates or appreciates, it would be a source of great shock to the parties entering into lending and borrowing business.

2. Money as store of value

It has been stated earlier that the introduction of money as a unit of account separates a single barter transaction implying simultaneous sale and purchase of equal value into two separate transactions of sale and purchase. This separation of a single barter transaction into two transactions enables one to act as a seller of one good at one place and time without being forced to act simultaneous as a buyer of another good. This separation, therefore, means separation both in time and place.

- ☐ Money as the medium of exchange has made it possible for people to sell goods and services at one place and time and to buy goods and services at another place and time. It is, however, not

possible to do so unless it is also possible to store the means of payment, i.e., purchasing power. It is, therefore, obvious that money necessarily acts as a temporary store of value by virtue of its use as the medium of exchange. In other words, in order to perform the function of medium of exchange, the value or purchasing power has to be stored in the form of money for temporary period in order to enable people to buy and sell at different times and places. Consequently, money has also to function as the temporary **abode of purchasing power** in order to function as medium of exchange in the economy.

As a store of value money is used to save purchasing power from the time income is received until the time it is spent.

Do you think money is the only asset used as a store of value? Why? Explain.

Money is not unique as a store of value. Any valuable asset may serve the purpose of storing the value. In fact, all assets like saving accounts, bonds, stocks, treasury bills, and real estate do serve as stores of value. As compared to money, these assets as store of value have some special advantages. They ordinarily yield an income in the form of interest, profits, rent or usefulness and they sometimes rise in value in terms of money.

However, money is the best as a store of value. The main reason for money to use as a store of value is its **liquidity**. All the assets are not as much liquid as money. Money does not have to be converted into anything else in order to make purchases. Other assets, however, involve transaction cost both one in time or cash charges when they are converted in to money.

In general, as a store of value, people's preference of money over other assets flows essentially from the characteristic **liquidity** of money and the uncertainty about the future value of nonmoney assets.

3. Money as transferor of value

By serving as a store of value money extends the marketing function over time. The use of purchasing power or value is transferred for one time to another. An equally important function of money is to help in the transference of value from one place to another and from one person to another.

- **Money serves both for the time-transfer and place-transfer of purchasing power.** Since money embodies value in its most general form, to transfer money is to transfer generic value for one

place and person to another place and person. It does not involve any difficulty to transfer the value of different assets and properties from one place to another within the country or between the two countries.

In general, since money is generally acceptable to people, a person can sell his property at one place and purchase property at another place. This also leads to the use of idle money lying with one person and thus it facilitates the lending and borrowing activities and also to its use for more productive purposes. In other words, the surplus capital at one place is invested at another place where it is needed. The **international flow of capital** has become possible mainly because money is used for the transfer of value.

III. Contingent functions of Money

Besides performing the primary and secondary functions, money also performs some more functions which David kindly has termed as contingent functions. These functions arise from the characteristics of the stage of economic life in which money is used. Therefore, Kinley has mentioned the following four contingent functions of money:

1. Distribution of Social income:- Money facilitates the distribution of social income or the national income between the different factors of production. A lightly specialized process of production at a large scale needs an active co-operation of millions of people as workers, land owners, capitalists, organizers and entrepreneurs. The resulting production belongs to all of them jointly. Without money, the distribution of the product among the various factors of production would be impossible. Money not only facilitates the distribution of social income, it also helps in bringing justice in distribution.

2. Equalization of marginal utilities and marginal productivities: - Money can be distributed in various competing uses in different quantities. It also helps in measuring the utility derived out of expenditure. A consumer gets maximum satisfaction when he adjusts his expenditure in such a way that each unit of money spent by him brings him goods of equal marginal utility. Money plays a significant role in equalizing these marginal utilities as the prices of all goods are expressed in terms of money.

- In the same manner, money enables the producer to maximize his returns by substituting one factor for another at the margin so that marginal productivity of each factor is equalized. In other words, to get more and more profits, commodities should be produced at minimum possible costs. In

order to accomplish this task, he combines the various factors of production in such a way as to equalize their respective marginal productivities. Money plays a decisive role in this context also as the productivities are ultimately measured and expressed in terms of money.

3. Basis of Credit system: - Money serves as a basis of the vast structure of modern credit system. Banks create credit on the basis of their cash reserves. All credit is based on money and is always expressed in terms of money. Credit is the foundation of modern commerce and it has played an important role in the development of modern economic system. Any change in the volume of credit is brought about mainly by an increase or a decrease in money supply.

4. Increases productivity of Capital: - Capital is found in various forms in the society. Among all forms of capital, money is the most liquid form of capital. That is, the moment an asset is converted into money, it becomes the most liquid and mobile asset. It confers on its possessor a social and economic power which attaches to no other form of wealth. It gives him general command over all saleable goods and services of others. Money flows in to the field of production as soon as it is needed there. **It is true that money is not a productive agent, but it cannot be denied at the same time that it helps in extending production.** It exercises an influence to deep production at its maximum. Since money gives a liquid form to wealth, many people prefer to store their wealth in the form of money rather than in non-money form.

? Please revise the different spectrums (types) of moneys regarding many criteria

1.2 Measurement of Money

In the past, money was the means of payment generally accepted in exchange, but it also had the characteristic that it did not pay interest. Thus the sum of currency and demand deposits (which did not earn interest) was the accepted definition of money for a long time.

Basically, there are four important theories that narrate the definition and measurement of money.

These are explained one by one in the section that follows:

1. Conventional approach
2. Chicago approach
3. Gurley and Shaw approach
4. Central bank approach

1. The Conventional Approach

This is the oldest approach. According to this the most important function of money in society is to act as a medium of exchange. Money is what it uniquely does. Keynes defined money as —that by delivery of which debt contracts and price contracts are discharged and in the shape of which general purchasing power is held. The types of assets that satisfy these criteria are

- a) Currency
- (c)

- b) Demand deposits in commercial banks (DD)

Thus according to the conventional definition

$M = C + DD$, this is the narrow definition of money. It excludes time deposits in the commercial banks because such deposits must first be converted in to either currency or demand deposits before they can be spent.

Note that demand deposits are deposits payable on demand through cheque or otherwise. It is important to note that among deposits it is only demand deposits which serve as a medium of exchange,

2. The Chicago approach

The Chicago economists led by Professor Milton Friedman adopted a broader definition of money by defining it more broadly as —**a temporary abode of purchasing power**. Their argument is that since in the economy money income and spending flow streams are not perfectly synchronized in time in order to function as a medium of exchange, money should be temporarily stored as a general purchasing power. This could be in the form of currency, demand deposits or time deposits (including saving deposits).

$$M2 = C + DD + SD + TD,$$

Where

C – Currency

SD – Saving
deposits

DD – Demand deposits

TD – Time deposits

(Checking deposits :)

The Chicago economist advanced two reasons for including time and saving deposits in the definition of money.

1. National income is more highly correlated with money that includes saving and time deposits than money narrowly defined.
2. According to economic theory, perfect or near perfect substitutes of a commodity should be included in the definition of a single commodity. According to Chicago economist time deposits (deposits which are not payable on demand and on which cheques can't be drawn) are very close substitutes for currency and demand deposits.

3. Gurley and Shaw approach

This approach is associated with the names of Professor John G. Gurley and Edwards Shaw. According to these economists there exists a fairly large spectrum of financial assets which are close substitutes for money. They emphasized the close substitution relationship between currency, demand deposits, commercial deposits, saving deposits, credit issued by credit institution, shares, government bonds and the like, all of which are regarded as alternative liquid stores of value by the public.

A rapid growth of deposits held by non – bank financial institutions (n.b.f.i.s) has increased their practical importance as a source of credit.

$M = C + DD + SD + TD + \text{non-clearing bank Deposits} + \text{n.b.f.i deposits.}$

The definition includes all deposits of and the claims of all types of financial intermediaries. It assigns weights to each asset in the definition of money to come up with the total supply of money. That is assignment of weights according to the degree of substitution.

Example: - Suppose the public's total assets consist of 200 Birr in the form of currency, 400 Birr worth of bank shares and 1000 Birr worth of ceiling fans. Thus, the public's total assets are worth 1600 Birr. It may be further assumed that the asset demands for currency and for ceiling fans are quite independent of one another while the degree of substitutability between the bank shares and currency is 0.50. **The weighted sum of the money supply would be equal to 400 Birr or 25%**

of the total assets because currency would be assigned a weight of one, bank shares a weight of one-half and the ceiling fans a weight of zero.

Theoretically this approach is superior to the Chicago which assigns equal weights to all items in the definition of money ranging from currency to time deposits. However practically it is difficult to implement it.

4. The central bank approach (Radcliff committee)

This approach, which has been favored by the commercial bank authorities, take the widest possible view of money. It defines money as

$M = C + DD + SD + TD + \text{non-clearing bank deposits} + \text{n.b.f.i deposits} + \text{credit lines}$. Money is identified with the credit extended by various sources. The reason for identifying

Money with credit used in the broadest possible sense of the term lies in the central banks historic position that —total credit availability constitutes the key variable for regulating the economy.

In general two pragmatic means could be used to define the money supply of a particular country.

1. The definition utilized should depend on the particular problem being studied. Example: if an analysis of the effect of the money supply on economic activity is being undertaken, the appropriate definition of money supply is the one that provides the best statistical results. If m_1 is statistically predictable than m_2 , monetary policy should be couched in terms of that narrow definition.
2. A method of identifying a break in the spectrum of assets to separate money. If the substitutability b/n DD and TD is lower than that between TD and other liquid assets, then the definition of money should be limited to currency and demand deposits.

1.3 Evolution of Money and the Payments System

Money was not discovered suddenly. It has undergone a process of evolution and this process has been continuous one. The evolution of money is divided in to several stages. These stages are

1. Barter system
2. Commodity money
3. Metallic Money
4. The system of coinage
5. Coins of inferior metals

6. Paper Money
7. Credit Money

1. Barter system: - In the beginning stages of human existence, trading was in the form of barter.

Barter means the exchange of goods for goods without the use of any medium of exchange. In other words; barter type of exchange was characterized by the absence of any medium of exchange. For example, the weaver exchanged his surplus cloth against the surplus corn of the farmer. There was, thus, no medium of exchange under barter system. However, the barter type of exchange is suffered from many difficulties. These difficulties may be analyzed as follows:

A. Lack of double co-incidence of wants: - No exchange can take place under barter unless the wants of two persons coincide. The double coincidence of wants and possession is a necessary condition for the exchange of commodities under the barter system. Two persons having exactly opposite and reciprocal wants should come together for the settlement of transaction under the barter type of exchange.

- In other words, barter requires a double coincidence of wants in the sense that for instance the owner of a quintal of teff who wants to exchange for a piece of cloth must find someone who both wants teff and possesses the cloth that the teff owner wants. He may find a person having cloth but that person may be in need of wheat. Thus, the number of men having something in surplus and wanting some other thing in exchange may multiply, but still no exchange may take place.

B. Difficulty of sub-division:- Another difficulty of barter is that all commodities cannot be sub-divided. Since the barter trade consists entirely of commodities without any medium of exchange, it poses a problem while exchanging a big article with any article of a smaller value.

Suppose a man has a cow and wants a chocolate for his son.

If he directly exchanges cow for a piece of chocolate, he is at a loss. The value of a cow is far greater than that of a chocolate. At the same time, it is not possible to divide the cow into a number of pieces without destroying its utility and value. Under such circumstances, exchange may not take place at all.

C. Difficulty of fixing price-ratios: - Every commodity is to be valued in terms of all other commodities under the barter type of exchange. As such every commodity will have as many price-ratios as the number of commodities available in the market. This creates the difficulty in

determining the price of a commodity. It is difficult to fix up the price-ratios for any commodity. For example, in a market where fifty commodities are available for exchange, every commodity will have 49 prices and every trader has to remember a number of price-ratios to carry the trade. Every time a new rate has to be fixed. This creates confusion on the conduct of transactions.

D. Difficulty of storing wealth: - There is also the difficulty of storing value or wealth under the barter system. Since the barter trade consists entirely of commodities, storing of wealth under barter would mean storing of commodities themselves. This is almost impossible physically. Most of the goods are perishable in nature and they cannot be easily stored for a long time.

E. Difficulty of deferred payments: - Another disadvantage of barter system is the lack of any satisfactory unit in terms of which contracts requiring future payments can be made. Contracts involving future payments are a part and parcel of an exchange economy. Individuals have to enter into agreements as regards to wages, salaries interest, rents and other such prices extending over a period of time. But in the barter type of exchange, these future payments are to be stated in terms of specific goods. This leads to three great difficulties.

2. **Commodity Money:** - Commodity money is considered to be the first form of money.

Many commodities came to be used as money. For sometimes, these commodities proved to be suitable for exchange. But with the expansion of the size of the market and increase in the number of commodities entering the trade, the commodity money was found to be unsatisfactory. The disadvantages of commodity money became apparent in course of time.

3. **Metallic Money:** - The metallic money could overcome all the difficulties of commodity money. Metals can be easily handled and their quantity can also be easily ascertained. Since they are not perishable, there is no difficulty in storing them. In the beginning, all types of metals were used as money. With the passage of time all other metals receded to the background barring the precious metals like gold and silver.

- Like other forms of money, the metallic money also suffered from a few disadvantages such as testing the quality of the metal every time, approaching the goldsmith or blacksmith for making smaller pieces involving wastage of time and money etc. So because of this difficulties human being introduce a new system called coinage.

4. **The system of coinage:** - The kings and the governments of the countries came forward to remove the disadvantages of the metallic money. They decided the standard size,

weight and quality of the Coin. They stamped their images on one side and the value of the coin on the other side. People accepted these coins due to their belief in the king or the government. With the passage of time coins came to be regarded as a matter of prestige by the king or the government.

5. **Coins of inferior Metals:** - Looking to the advantages of the system of coinage, people developed excess faith in the king or the government. Taking the advantage of the faith of people, kings and the governments started manufacturing coins of inferior metals. Here coins of inferior metals indicate coins made of metals other than gold and silver. Nevertheless, the values of the coins were maintained. Governments took the monopoly of issuing the coins and thus the coins made of inferior metals came to be accepted by people. Here we find a change from the full-bodied money to token coins.

6. **Paper Money:** - Paper money is of recent origin. It was invented to overcome the difficulties faced by the system of coinage. Cons proved to be suitable only for concluding small transactions. It was not possible to carry on large transactions with coins. It also posed the problem of transferring the coins from one place to another. Paper-notes could overcome these difficulties. These notes are issued by the central bank of the country.

7. **Credit Money:** - Transactions involving a huge amount of money are made easier by credit money. In fact, credit money came into vogue for this purpose. Paper-notes are not always safe and suitable to be carried from one place to another. Credit money overcomes this difficulty. Banks in a country are entrusted with the task of issuing credit money in different forms of negotiable instruments like checks, drafts, bills of exchange, promissory notes etc. With the discovery of credit money, it has become possible and easier to conclude the transactions of a higher order and the transactions among different countries.

1.4 Monetary theory

The classical monetary theory essentially was an extension of classical value theory, with no doctrinal dichotomy what so ever. Since money was a commodity in classical analysis, therefore its value was simply explained by the same value theory that explained the prices of other commodities.

The classical monetary theory concluded, *ceteris Paribus*, the value of the money depended inversely and proportionally on the supply of money. This was actually what the classical quantity theory reflected. Accordingly, the doubling of the money supply, keeping all else constant would double the price level (i.e. halve the value of money). In short, money in classical analysis was explained just like any other commodity, being distinguished from other commodities only by **the unitary elasticity of its demand function** and by the **peculiarity of its supply**

- Neo-classical monetary theorists are the Anglo Americans who professed to interpret the classical monetary theory but inadvertently turned it upside down, without even realizing or detecting the inversion they caused to the doctrine. The commodity and quantity theories of money complemented each other in explaining the value of money in classical monetary analysis; however these theories were presented as competing theories in neo-classical analysis, interpreted as half-truths.

In conclusion, money remains the most talked about and discussed entity of the historic times; however its value determination theory took a roller coaster ride from Classical to Neo Classical eras. Classical monetary economists rendered value determination purely based on cost of production in the long run and by the forces of demand and supply in the short run.

Neo-Classicals in an attempt to interpret the classical monetary thought turned it upside down by extending the short run application of Quantity theory to the long run and excising its actual applicability.

Neo-classical monetary theory remains essentially a quantity theory in a long-run context and this is the theory that dominates the mainstream economics now a days. In contrast to mainstream monetary economics a heterodox school of economic thought, the Austrian School considers the inelastic nature of gold and commodity money, a virtue and not a vice. The economists of this school explain how trade cycles are exacerbated by the extension of endless credit. They propose the return to the gold standard.

Chapter Two

An overview of the Financial System and Interest Rates

2.1. Functions and structures of Financial Markets

Motivation on General Terms and Concepts

Asset: An asset is anything of durable value, that is, anything that acts as a means to store value over time.

Real assets: These are assets in physical form (e.g., land, equipment, houses..), including "human capital" assets embodied in people (natural abilities, learned skills, knowledge,..).

Financial assets are claims against real assets, either directly (e.g., stock share equity claims) or indirectly (e.g., money holdings, or claims to future income streams that originate ultimately from real assets).

Securities are financial assets exchanged in auction and over-the-counter markets whose distribution is subject to legal requirements and restrictions (e.g., information disclosure requirements).

- ☐ At any point in time in an economy, there are individuals or organizations with **excess amounts of funds**, and others with a **lack of funds** they need for example to consume or to invest.
- ☐ Exchange between these **two groups of agents** is settled in **financial markets**

The first group is commonly referred to as **lenders**, the second group is commonly referred to as the **borrowers** of funds.

Lenders: Are people who have available funds in excess of their desired expenditures that they are attempting to loan out.

Borrowers: Are people who have a shortage of funds relative to their desired expenditures who are seeking to obtain loans. Borrowers attempt to obtain funds from lenders by selling to lenders newly issued claims against the borrowers' real assets, i.e., by selling the lenders newly issued financial assets.

□ A **financial market** is a market in which financial assets are traded (bought and sold). In addition to enabling exchange of previously issued financial assets, financial markets facilitate borrowing and lending by facilitating the sale by newly issued financial assets.

✎ **Financial market** refers to an elaborate system of the financial institution and intermediaries and arrangement put in place and developed to facilitate the transfer of funds from surplus economic units (savers) to deficit economic units (investors).

✎ Therefore, financial markets are institutions or arrangements that facilitate the exchange of financial assets, including deposits and loans, corporate stocks and bonds, government bonds and more exotic instrument such as options and futures contracts.

A. According to type of assets: debt market vs. equity market

A Firm or an individual can obtain funds in a financial markets in two ways.

Debt Market: The debt market is the market where debt instruments are traded. Debt instruments are assets that require a fixed payment to the holder, usually with interest. Examples of debt instruments include bonds (government or corporate) and mortgages.

The maturity of a debt instrument is the number of years or terms until the instrument's expiration date. A debt instrument is short-term if it is less than one year, long term if its maturity is ten years or longer.

Equity Market: The equity market (often referred to as the stock market) is the market for trading equity instruments. Stocks are securities that are a claim on the earnings and assets of a corporation. An example of an equity instrument would be common stock shares, such as those traded on the New York Stock Exchange.

B. According to maturity of assets: Money versus Capital Markets:

Money market is the sector of the financial market that includes financial instruments that have a maturity or redemption date that is one year or less at the time of issuance. These are mainly **wholesale markets**. *Examples:* Treasury bills; negotiable bank certificates of deposit (CDs); commercial paper; Eurodollars/Eurocurrency market etc.

The **capital market** is the sector of the financial market where long-term financial instruments issued by corporations and governments trade. Here —long-term‖ refers to a financial instrument with an original maturity greater than one year and perpetual securities (those with no maturity). There are two types of capital market securities: those that represent shares of ownership interest, also called **equity**, issued by corporations, and those that represent indebtedness, or **debt** issued by corporations and by the state and county governments. This includes corporate bonds, debentures and equity, treasury bonds and the like. C. **Delivery of assets: spot market and Forward Market**

This classification applies to the foreign exchange market. **Cash/spot market:** asset exchange immediately after trade (2 days) and derivative market: asset exchange occurs on future date after trade. The **cash market**, also referred to as the **spot market**, is the market for the immediate purchase and sale of a financial instrument.

In contrast, some financial instruments are contracts that specify that the contract holder has either the obligation or the choice to buy or sell another something at or by some future date. The —something‖ that is the subject of the contract is called the *underlying (asset)*. The underlying asset is a stock, a bond, a financial index, an interest rate, a currency, or a commodity.

D. Primary versus Secondary Markets



Primary markets are securities markets in which newly issued securities such as bonds or shares are offered for sale to buyers. The new issues of financial instruments can be through private placement, initial public offers, book building and seasoned public offers. Primary markets

are meant to raise capital for corporations and government. The initial issuer raises funds only through the primary market. An important institution that assists in the primary market is the investment banks. It does this by underwriting the initial public offer.

 Secondary markets are securities markets in which existing securities that have previously been issued are resold. Secondary markets are important for two reasons: first they make it easier for investors to buy or sell financial securities to raise cash, that they make the financial security more liquid. Secondly they assist in price determination of the financial securities.

E. Structure of market: Organized exchange, over the counter market

A. **Organized exchange:** buyers and sellers meet in physical location to trade. Specifically, organized exchanges permit buyers and sellers to trade with each other in a centralized location, like an auction. However, securities are traded on the floor of the exchange with the help of *specialist traders* who combine broker and dealer functions. The specialists broker trades but also stand ready to buy and sell stocks from personal inventories if buy and sell orders do not match up.

B. **Over-the-counter (OTC) market:** An *over-the-counter market* has no centralized mechanism or facility for trading. Instead, the market is a public market consisting of a number of dealers spread across a region, a country, or indeed the world, who *make the market* in some type of asset. That is, the dealers themselves post bid and asked prices for this asset and then stand ready to buy or sell units of this asset with anyone who chooses to trade at these posted prices. **Over the Counter Market (OTC)** Provides an opportunity for unlisted/unquoted firms to sell their security.

2.2 Financial Instruments

Financial instruments (securities or assets) are of two types: primary securities and secondary securities. **Primary securities (like bonds and mortgages for example) are issued directly by the ultimate borrowers**, while secondary securities are issued by financial institutions. Besides, primary securities are claims against real sector units while secondary securities are financial claims issued by financial institutions or intermediaries against themselves to raise funds from the public. Even, based on time period, financial instruments may be divided into two:

Money market instruments- these are short term instruments with duration of a year or less. They are used to raise short term funds to buy goods and services in the market. **Capital market instruments**- these are medium term and long term instruments. The basic productivity capacity is embodied in these instruments. Good examples include land and equipment.

Money Market Instruments (short term instruments)

Because of their short terms to maturity, their prices are relatively stable and are the least risky investments. There are two common kinds of money market instruments (although their definitions depend on the structure of the financial market): Bills of exchange and Inter- Bank Call Money.

Bills: - these are in turn divided into two: Treasury Bills (government bills) and Commercial Bills.

1. **Treasury bills:** these refer to financial instruments that the government (specifically the ministry of finance) sells in the market to collect money. Governments are tempted to use it whenever there is shortage of money for a short period of time (normally a maximum of 6 months). They are sold on discount basis; and there is no interest payment to be made on them. They are negotiable.
2. **Commercial Bill:** It is one which arises out of a genuine trade transaction, i.e. credit transaction. As soon as goods are sold on credit, the seller draws a bill on the buyer for the amount due. The buyer accepts it immediately agreeing to pay the amount mentioned therein after a certain specified date.

□ Thus, a bill of exchange contains a written order from the creditor to the debtor, to pay a certain sum, to a certain person, after a creation period. A bill of exchange is a —self-liquidating paper and negotiable/it is drawn always for a short period ranging between 3 months and 6 months.

□ Commercial bills are three- party bills in the sense that a drawer, a drawee and a commercial bank sign on the bill.

Inter- Bank Call Money: - this refers to the transactions made among banks due to short term deficit and/or surplus. It is an instrument that is transacted in a market among banks where surplus banks are ready to provide liquidity to the deficit banks. It is an accommodation made by banks to one another's requirement. It is an over- night cash movement among banks activated by brokers. The brokers (financial mediators) have good deal of information about the liquidity position of each bank and every day. The banks are assumed not to have information on other banks' (rival banks) liquidity position [otherwise the brokers will have no reason to intervene in

the market]. The banks make transactions based on interest rates- called *Call Rate*. Among all interest rate types, *call rates* are the most volatile ones.

□ The main function of the call money market is to redistribute the pool of day – to – day surplus funds of banks among other banks in temporally deficit of cash position. For smooth and efficient functioning of the banking system, there must be some institutional arrangements for making the surplus of some banks available to others in temporally deficits. Hence using the call money market, banks will economize their cash and improve their liquidity position. The call money market is highly competitive and sensitive market. It registers very quickly the pressure of demand of and supply for funds operating in the money market.

Capital Market Instruments (long term instruments)

✎ Precisely speaking, capital market instruments are debt and equity instruments with maturities of over one year. They have far wider price fluctuations than money market instruments, and are considered to be fairly risky investments.

✎ A good example is found in the market for stocks, corporate bonds, and mortgages. Stocks are equity claims on the net income and assets of a corporation. Corporate bonds are long-term bonds issued by corporations with very strong credit ratings. Mortgages are loans to individuals or business firms to purchase buildings, land or other real assets where the structure of the asset serves as collateral for the loan.

2.3. Financial intermediaries

Financial intermediaries are financial institutions operating in a money market. They are the link between the borrowers and the lenders. They deal in securities primary as well as secondary.

There are two types of financial intermediaries:

- A. Banking institutions and
- B. Non banking institutions

Since financial intermediaries are the line between the borrowing individuals and firms, they have to satisfy the portfolio preferences of both the parties. The borrowers produce and purchase current output. Their interest is to expand their holdings of real assets like real estate, plants and equipments, for which they issue primary securities. These bonds, equities, debts, mortgages, bills are their liabilities.

 The lenders, on the other hand, possess assets in the form of bank deposits, insurance policies pensions, etc. The financial intermediaries transfer the savings of lenders (also known as surplus units) to the borrowers also known as deficit units) by purchasing primary securities and issuing secondary securities.

 These secondary securities are the currency issued by the central bank, demand and time deposits of commercial banks, savings deposits, insurance and pension funds of non-monetary intermediaries both the primary and secondary securities are known as financial assets. When the primary securities are purchased directly by surplus income units, it is called as direct financing and when these are purchased through financial intermediaries then it is called as indirect financing.

A. Banking financial intermediaries (BFIs):- As already stated in the previous discussions banks are the most important financial intermediary in the monetary system. For example, the intermediation by the commercial banks means the ultimate borrowers sell their primary securities to these banks and receive money in the form of demand deposits in these banks. The demand deposits are mostly spent by the borrowers to produce and purchase current output. The lenders acquire financial assets from the borrowers through bank deposits. Thus, the ultimate borrowers have created primary securities, banks have created primary securities, banks have created money by purchasing them and the ultimate lenders have acquired financial assets as a compensation for not spending. The bank intermediation has therefore, led to the transfer of unspent incomes from the surplus to deficit units.

B. Non-bank financial intermediaries (BNFIs):- The non-bank financial intermediaries refer to those financial institutions, other than the banking companies, which are engaged in some form of borrowing and lending activities. These include the savings banks, building societies, hire purchase companies. Insurance companies, and the investment and unit trusts.

Regarding to the intermediation by the non-banking financial intermediaries, ultimate lenders issue checks and present them to these financial institutions making a demand on their deposits with commercial banks. These institutions endorse the checks and send them to the commercial banks. The commercial banks purchase primary securities from ultimate borrowers who now have demand deposits in them which they spend for the purchase of current output and are ultimate

received by the financial institutions have created secondary securities and the lenders have acquired financial assets.

□ This process only involves the transfer of bank deposits from the ultimate lenders to the non-banking financial institutions then to ultimate borrowers and finally back to ultimate lenders. The banks have not intermediated in this process and their role has been confined only to that of administering the transfer of demand deposits in their ledgers.

Functions of financial intermediaries

Financial markets attract and allocate savings (1) investment and (2) financing function, (3) pricing function, (4) payment function and (5) risk management (6) Asymmetric information problem resolution.

Intermediary Functions: The intermediary functions of a financial markets include the following:

- **Transfer of Resources:** Financial markets facilitate the transfer of real economic resources from lenders to ultimate borrowers. Financial markets channel funds from households, firms, governments and foreigners that have saved surplus funds to those who encounter a shortage of funds (for purposes of consumption and investment). They allow funds to move from people who lack productive investment opportunities to people who have such opportunities.
- **Enhancing income:** Financial markets allow lenders to earn interest or dividend on their surplus funds, thus contributing to the enhancement of the individual and the national income.
- **Risk Sharing and Distribution** Risk sharing : financial markets allow transfer of risk from those undertaking investments to those who provide funds for those investments
- **Productive usage:** Financial markets allow for the productive use of the funds borrowed. The enhancing the income and the gross national production. ○ **Capital Formation:** Financial markets provide a channel through which new savings flow to aid capital formation of a country.
- **Price determination:** Financial markets allow for the determination of price of the traded financial assets through the interaction of buyers and sellers. They provide a sign for the allocation

of funds in the economy based on the demand and supply through the mechanism called price discovery process.

- **Sale Mechanism:** Financial markets provide a mechanism for selling of a financial asset by an investor so as to offer the benefit of marketability and liquidity of such assets.
- **Information:** The activities of the participants in the financial market result in the generation and the consequent dissemination of information to the various segments of the market. So as to reduce the cost of transaction of financial assets.
- **Financial Efficiency** Financial markets reduce information asymmetry and transaction costs. The facilitation of financial transactions through financial markets lead to a decrease in informational cost and transaction costs, which from an

□ They can reduce the chances of liquidity crises by careful scheduling of the maturities of primary securities. In addition they also get those tax benefits which are not available to individual lenders. On the borrowing side, the number of depositors being very large, the financial intermediaries can normally set a firm schedule of claims for repayment to their depositors. It is in this way that the financial intermediaries encourage saving and investment and promote economic growth.

- The financial intermediaries with their ample resources can thinly spread their risks among different borrowers.
- All financial intermediaries create financial assets and thus substantially add to the stock of financial assets available to the lenders.
- All financial intermediaries provide liquidity through conversion of an asset in to cash easily and quickly without any loss in their value in terms of money.
- Since financial intermediaries compete with each other their savings with the financial intermediaries which invest them in primary securities.

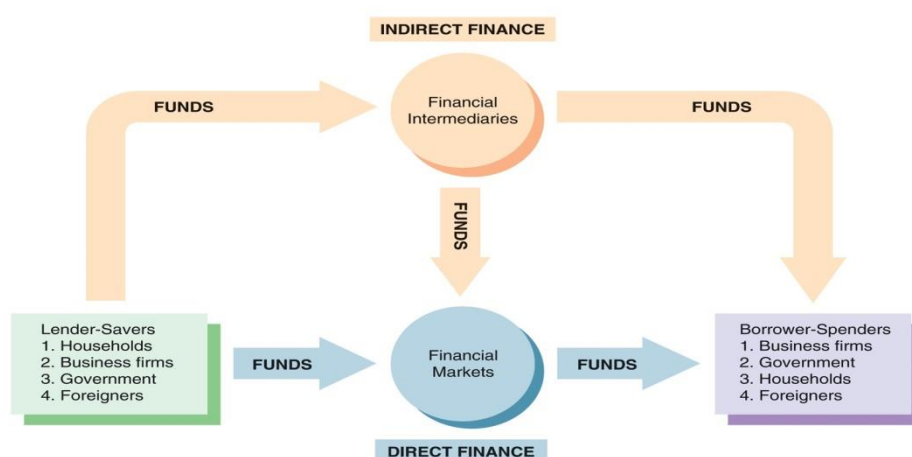
The prices of securities are bid up and interest rates fall low rates of interest are to the advantage of both the investors and the savers.

By maintaining equilibrium between the demand and supply of financial assets the financial intermediaries bring about stability in the capital market.

Forms of Exchange in Financial Markets

Financial markets perform the essential function of channeling funds from households, firms, government that have surplus funds to those who have a deficit of funds. For consumption and investment. The most important savers (lenders) are households, business firms, governments and foreigners. Important borrowers are households, business enterprises and government. They allow funds to move from people who lack productive investment opportunities to people who have such opportunities. Financial markets are crucial for producing an efficient allocation of capital. Which contribute to higher production and efficiency for the overall economy.

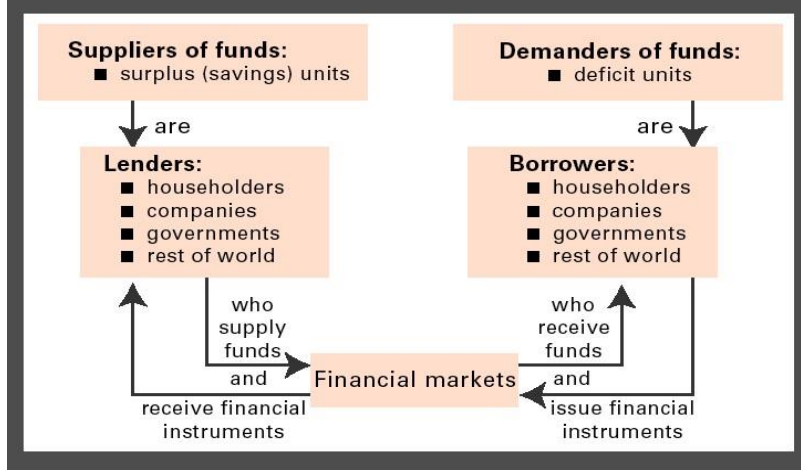
Financial markets operating efficiently improve the economic welfare of all members of the society. Funds flow through the financial market through two channels, as illustrated in figure



1. Direct Finance

- Borrowers borrow directly from lenders in financial markets by selling financial instruments which are claims on the borrower's future income or assets. The financial security or instruments are assets to the savers (lenders) but a liability to the individual and firms who sell or issue them. Financial security or instruments are assets to the savers (lenders) but a liability to the individual and firms who sell or issue them.

Figure 1.1 Financial markets and flow of funds relationship



2. Indirect Finance

□ Borrowers borrow indirectly from lenders via financial intermediaries (established to source both loanable funds and loan opportunities) by issuing financial instruments which are claims on the borrower's future income or assets

□ The process of indirect finance is known as financial intermediation. It involves the use of financial intermediaries. Financial intermediaries are important because they match sellers and buyers **indirectly** through the process of **financial asset transformation**. Financial intermediation is the primary route for moving funds from borrowers to lenders.

2.4. Interest rates and their measurement

✎ Interest rates are among the most closely watched variables in the economy. Their movements are reported almost daily by the news media, because they directly affect our everyday lives and have important consequences for the health of the economy.

✎ They affect personal decisions such as whether to consume or save, whether to buy a house, and whether to purchase bonds or put funds into a savings account. Interest rates also affect the economic decisions of businesses and households, such as whether to use their funds to invest in new equipment for factories or to save their money in a bank.

✎ The *yield to maturity* is the most accurate measure of interest rates; the yield to maturity is what economists mean when they use the term *interest rate*.

- The concept of **present value** (or **present discounted value**) is based on the commonsense notion that a dollar paid to you one year from now is less valuable to you than a dollar paid to you today: This notion is true because you can deposit a dollar in a savings account that earns interest and have more than a dollar in one year.
- If an individual has an initial amount P_0 which he/she lends at market rate of interest „ i “, then he/she will have P_1 at the end of one year. Now again he/she lends his initial amount of P_1 at „ i “ he/she will get P_2 amount at the end of two years and so on. Mathematically, this can be represented as follows:

$$\text{First year} \Rightarrow P_1 = P_0 + iP_0 = P_0(1+i)^1 \dots\dots\dots(1)$$

$$\text{Second year} \Rightarrow P_2 = P_1 + iP_1 = P_1(1+i)^1 \dots\dots\dots(2)$$

Since $P_1 = P_0(1+i)$, by substituting equation (1) in equation (2) we get the following result:

$$\begin{aligned} P_2 &= P_0(1+i)(1+i) \\ P_2 &= P_0(1+i)^2 \dots\dots\dots(3) \end{aligned}$$

Similarly, the person again invests for another year at the market rate of interest, and then he/she would have $P_0(1+i)^3$ at the end of the third year. Following this, we can have a general formula which can be written as:

$$P_n = P_0(1+i)^n \dots\dots\dots(4)$$

Where P_n is the amount one gets at the end of 'n' years at market rate of interest (i) if one invests P_0 amount today or this year. The present value of income to be received after 'n' years equals the amount an individual would have to lend at the market rate of interest 'i' for 'n' years in order to receive the given amount P_n . The present value given by ' P_0 ' is obtained by solving for ' P_0 ' from equation (4) above. The general formula for the present value of future income is given as follows:

$$P_0 = \frac{P_n}{(1+i)^n} \dots\dots\dots(5)$$

□ We discount every year and add the present value of each income received in future. After discounting every year we can obtain the general formula of present value (PV). Which is:

$$PV = \frac{P_1}{(1+i)^1} + \frac{P_2}{(1+i)^2} + \frac{P_3}{(1+i)^3} + \dots\dots + \frac{P_n}{(1+i)^n} \dots\dots\dots(6)$$

✎ To understand this concept let us take an example. Suppose an individual is to receive 1,100 Birr after one year from today. If the market rate of interest is 10 percent, what is

the value of 1,100 Birr today?

For example, if you made your friend, a simple loan of \$100 for one year, you would require her to repay the principal of \$100 in one year's time along with an additional payment for interest; say, \$10. In the case of a simple loan like this one, the interest payment divided by the amount of the loan is a natural and sensible way to measure the interest rate. This measure is the so-called *simple interest rate*, i , is:

$$i = \frac{\$10}{\$100} = 0.10 = 10\%$$

If you make this \$100 loan, at the end of the year you would have \$110, which can be rewritten as:

$$\$100 \times (1 + 0.10) = \$110$$

If you then lent out the \$110, at the end of the second year you would have:

$$\$110 \times (1 + 0.10) = \$121$$

or, equivalently,

$$\$100 \times (1 + 0.10) \times (1 + 0.10) = \$100 \times (1 + 0.10)^2 = \$121$$

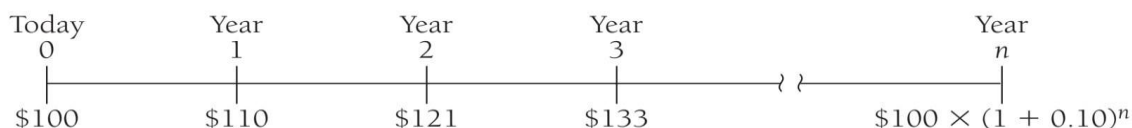
Continuing with the loan again, you would have at the end of the third year:

$$\$121 \times (1 + 0.10) = \$100 \times (1 + 0.10)^3 = \$133$$

Generalizing, we can see that at the end of n years, your \$100 would turn into:

$$\$100 \times (1 + i)^n$$

The amounts you would have at the end of each year by making the \$100 loan today can be seen in the following timeline:



This timeline immediately tells you that you are just as happy having \$100 today as having \$110 a year from now (of course, as long as you are sure that Jane will pay you back). Or that you are just as happy having \$100 today as having \$121 two years from now, or \$133 three years from now or $\$100 \times (1 + 0.10)^n$, n years from now. The timeline tells us that we can also work backward from future amounts to the present: for example, $\$133 = \$100 \times (1 + 0.10)^3$ three years from now is worth \$100 today, so that:

$$\$100 = \frac{\$133}{(1 + 0.10)^3}$$

The process of calculating today's value of dollars received in the future, as we have done above, is called *discounting the future*. We can generalize this process by writing

today's (present) value of \$100 as PV , the future value of \$133 as FV , and replacing 0.10 (the 10% interest rate) by i . This leads to the following formula:

$$PV = \frac{FV}{(1 + i)^n}$$

Nominal versus Real interest rate

The **nominal interest rate** (or **money interest rate**) is the percentage increase in money you pay the lender for the use of the money you borrowed. For instance, imagine that you borrowed \$100 from a bank one year ago at 8% interest on your loan. When you repay the loan, you must repay the \$100 you borrowed plus \$8 in interest—a total of \$108.

But the nominal interest rate doesn't take inflation into account. In other words, it is unadjusted for inflation. To continue our scenario, suppose on your way to the bank a newspaper headline caught your eye stating: —Inflation at 5% This Year!! Inflation is a rise in the general price level. A 5% inflation rate means that an average basket of goods you purchased this year is 5% more expensive when compared to last year. This leads to the concept of the **real**, or inflationadjusted, **interest rate**.

□ The real interest rate measures the percentage increase in purchasing power the lender receives when the borrower repays the loan with interest. In our earlier example, the lender earned 8% or \$8 on the \$100 loan. However, because inflation was 5% over the same time period, the lender actually earned only 3% in real purchasing power or \$3 on the \$100 loan

Chapter Three

The demand for Money and other assets

3.1. Quantity theory of money

The quantity theory of money is a theory that discusses how the nominal value of aggregate income is determined. It is also a theory of money demand since it tells how much money is held for a given amount of aggregate income. The theory is based on the concept of velocity of money.

Velocity of Money: Is the rate of turnover of money. It is the average number of times that a unit of currency is spent in buying the total amount of goods and services produced in the economy. It provides the link between money supply and nominal income. Mathematically, it can be given as follows:

We first start from an identity which equates total spending with nominal income or nominal value of goods and services. The identity holds true by definition.

$$MV = PY \dots\dots\dots (1)$$

Where: V is velocity of money

M is money supply

P is average price level

Y is real level of income or volume of transaction and

PY is nominal income. From the above identity we derive the equation for velocity of money as follows:

$$V = \frac{PY}{M} \dots\dots\dots (2)$$

 According to classicalists, velocity of money depends on institutional and technological factors related to the paying behavior of people. It is therefore assumed to be constant in the short run since institutions and technologies do not change in the short-run. This implies that any change in money supply (M) will be reflected in the nominal income (PY) i.e., if money supply doubles so will, nominal income. Classicalists also assume that the aggregate real income (Y) will remain constant at the equilibrium level in the short-run since prices and wages are completely flexible. Thus, changes in money supply will lead to changes in price only. In other words, an increase in the supply of money leads to an increase in average price level, but everything else (real income, velocity of money etc) will not be affected.

For example, if V and Y are fixed at 4 and 100 respectively, a supply of money (M) which equals 25 implies that the average price level (P) should be 1. And if M doubles to 50 P will also double to 2 while everything else will remain unchanged. This is called the **quantity theory of money** which implies that money is **neutral with respect to the real sector** of the economy and movements in the price level results solely from **changes in the quantity of money**.

From the formula of velocity of money given above we can derive the formula for money holdings or money supply (M) as follows:

$$M = \frac{1}{V} PY \dots\dots\dots (3)$$

 In equilibrium, the amount of money that people hold (M) will be equal with the quantity of money people demand (quantity of money demand). Thus, we can substitute money demand for money supply in the equation for money supply and then derive the function for money demand. In effect, the quantity theory of money tells us how much money is people hold for a given amount of aggregate income and is a theory of demand for money.

$$\begin{aligned} M^d &= \frac{1}{V} PY \\ &= kPY \dots\dots\dots (4) \end{aligned}$$

Where M^d is quantity of money demanded and k is the reciprocal of velocity while the other symbols are defined as above. This suggests that money demand is purely a function of nominal income and is not affected by interest rate.

 Thus, the classicalists' theory of money demand shows that money demand is not a function of interest rate and this is backed by the assumption that people demand money only **for transaction purpose**. For classicalists, money demanded for transaction purpose only depends on the **volume of transaction** and does not depend on interest rate. The demand function also tells us that quantity of money demanded is a constant function of nominal income, i.e, and people hold a fraction of their nominal income in the form of money.

3.2 Keynes's Liquidity Preference Theory

Keynes developed a theory of demand that emphasized the importance of interest rates like the Cambridge economists. But, his approach is more detailed to the extent that he identified the different motives why people demand money and showed the effect of interest rate on the demand for money in a more explicit way. According to his theory, there are three motives for demanding

money namely: transaction motive, precautionary motive and speculative motive. We will first see the three motives briefly and then the function of demand for money.

Transaction Motive: This motive of demanding for money refers to the act of demanding money for the purpose of transaction. People demand money to fulfill their current consumption demand of goods and services i.e., they demand money as a medium of exchange. This part of the demand for money is determined by the amount of current consumption which according to Keynes is proportional to income; it is purely a function of income and interest rate does not affect it.

Precautionary Motive: Like the transaction motive this motive also refers to demand for money for the purpose of transaction. But, unlike the first motive, here money is not demanded for current transactions and it is rather demanded a cushion against **unexpected transactions in the future**. People could not predict their future volume of transactions and hence their future demand for money and they hold some amount of money as a precaution for future unexpected demand for money and this what Keynes called the precautionary motive for money demand.

People for example, may not be sure about their future health expense and may hold some money for fear they may face some health problem unexpectedly and need money for treatment. Another example could be expectation about price changes; they may suspect that the prices for some commodities may fall in the future and this will make them hold money so that they will purchase these commodities at better price.

For Keynes, this part of the demand for money is primarily dependent on the **volume of future** expected transactions which in turn is proportional to income like the transaction motive, i.e. as income increases, people will hold more money for both purposes-current transaction and future expected transactions.

Speculative Motive

Keynes agreed with the Cambridge economists that money is also demanded as a **store of value** (wealth) and he called this motive of holding money the **speculative motive**. Though this motive is affected by wealth which in turn is believed to be closely related with income Keynes also identified interest rate to be an important determinant of this part of money demand. To put it differently, people hold their wealth in different assets and money is one alternative of holding wealth. Therefore, as their wealth increases people will hold more of different assets including

money but for the same wealth people may make substitutions between assets based on the relative desirable properties of the assets like expected return.

Keynes considered money and bond as the two alternative assets of holding money and hence people can hold their wealth either in the form of money or bond and since money does not earn interest people will choose to hold bond if they expect a positive return from holding bond. Therefore, the **speculative motive** is inversely affected by interest rate, i.e., as interest rate from bonds increases the return from bonds will increase and people will shift from money to bonds (they will hold less money and hold bonds).

Though Keynes considered only bond as an alternative asset to money in terms of storing wealth, the conclusion can be generalized to the case where there are many alternative assets of storing wealth including different physical assets and financial assets other than bond. This part of the demand for money is therefore affected by **interest rate** and since the total demand for money is according to Keynes is a function of the three motives and it will also be affected by interest rate.

3.3. Post Keynesians developments in monetary theory

The main drawback of Keynes's speculative demand for money is that it visualizes that people hold their assets in either all money or all bonds. This seems quite unrealistic as individuals hold their financial wealth in some combination of both money and bonds.

□ This gave rise to portfolio approach to demand for money put forward by Tobin, Baumol and Friedman. The portfolio of wealth consists of money, interest-bearing bonds, shares, physical assets etc. Further, while according to Keynes's theory, demand for money for transaction purposes is insensitive to interest rate, the modern theories of money demand put forward by Baumol and Tobin show that money held for transaction purposes is interest elastic. According to him, an investor is faced with a problem of what proportion of his portfolio of financial assets he should keep in the form of money (which earns no interest) and interest bearing bonds. The portfolio of individuals may also consist of more risky assets such as shares.

According to Tobin, faced with various safe and risky assets, individuals diversify their portfolio by holding a balanced combination of safe and risky assets.

According to Tobin, individual's behaviour shows risk aversion. That is, they prefer less risk to more risk at a given rate of return. In the Keynes's analysis an individual holds his **wealth in either all money or all bonds depending upon his estimate of the future rate of interest.**

But, according to Tobin, individuals are uncertain about future rate of interest. If a wealth holder chooses to hold a greater proportion of risky assets such as bonds in his portfolio, he will be earning a high average return but will bear a higher degree of risk. Tobin argues that a risk averter will not opt for such a portfolio with all risky bonds or a greater proportion of them.

On the other hand, a person who, in his portfolio of wealth, holds only safe and riskless assets such as money (in the form of currency and demand deposits in banks) he will be taking almost zero risk but will also be having no return and as a result there will be no growth of his wealth. Therefore, people generally prefer a mixed diversified portfolio of money, bonds and shares, with each person opting for a little different balance between riskiness and return.



It is important to note that a person will be unwilling to hold all risky assets such as bonds unless he obtains a higher average return on them. In view of the desire of individuals to have both safety and reasonable return, they strike a balance between them and hold a mixed and balanced portfolio consisting of money (which is a safe and riskless asset) and risky assets such as bonds and shares though this balance or mix varies between various individuals depending on their attitude towards risk and hence their trade-off between risk and return.



The Baumol-Tobin model normally analyzes the costs and benefits of holding money. The benefit of holding money is convenience: people hold money to avoid making a trip to the bank every time they wish to buy something. The cost of this convenience is the forgone interest they would have received had they left the money deposited in a savings account that paid interest.

A Baumol analysis the transactions demand for money of an individual who receives income at a specified interval, say every month, and spends it gradually at a steady rate. This is illustrated in Fig. 3.1. It is assumed that individual is paid Birr 12000 salary cheque on the first day of each month.

Suppose he gets it cashed (i.e. converted into money) on the very first day and gradually spends it daily throughout the month. (Birr 400 per day) so that at the end of the month he is left with no money. It can be easily seen that his average money holding in the month will be Birr = $12000/2 = \text{Birr } 6,000$ (before 15th of a month he will be having more than Birr 6,000 and after 15th day he will have less than Birr 6,000).

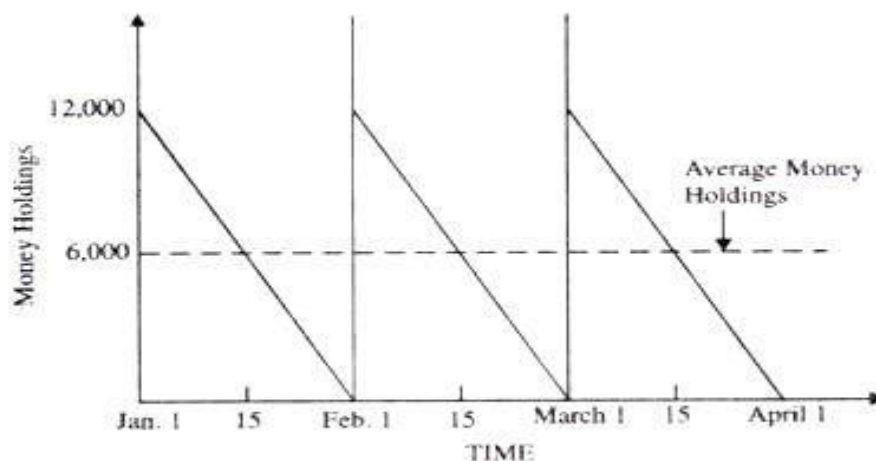


Fig. 19.2. Stream of Cash Payments and Transactions Demand for Money

Average holding of money equal to Birr 6,000 has been shown by the dotted line. Now, the question arises whether it is the optimal strategy of managing money or what is called **optimal cash management**. The simple answer is no. This is because the individual is losing interest which he could have earned if he had deposited some funds in interest-bearing saving deposits instead of withdrawing all his salary in cash on the first day.

He can manage his money balances so as to earn some interest income as well. Suppose, instead of withdrawing his entire salary on the first day of a month, he withdraws only half of it i.e. (Birr 6,000 in cash and deposits the remaining amount of Birr. 6,000 in saving account which gives him interest of say, 5 per cent, his expenditure per day remaining constant at Birr 400.

This is illustrated in Fig. 19.3. It will be seen that his money holdings of Birr 6,000 will be reduced to zero at the end of the 15th day of each month. Now, he can withdraw Birr 6,000 on the morning of 16th of each month and then spends it gradually, at a steady rate of 400 per day for the next 15 days of a month. This is a better method of managing funds as he will be earning interest on Birr 6,000 for 15 days in each month. Average money holdings in this money management scheme is $\text{Birr } 6000/2 = 3000$

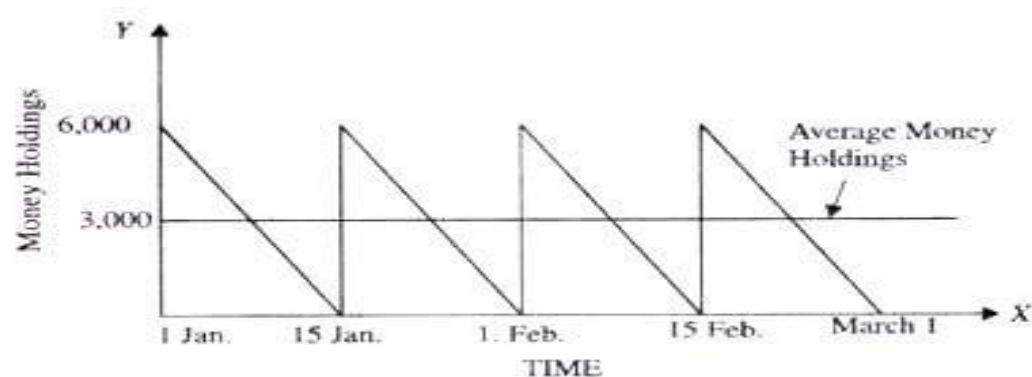


Fig. 19.3. Transactions Demand for Money and Stream of Cash Payments

Likewise, the individual may decide to withdraw Birr 4,000 (i.e., 1/3rd of his salary) on the first day of each month and deposits Birr 8,000 in the saving deposits. His Birr 4,000 will be reduced to zero, as he spends his money on transactions, (that is, buying of goods and services) at the end of the 10th day (400×10) and on the morning of 11th of each month he again withdraws Birr 4,000 to spend on goods and services till the end of the 20th day and on 21st day of the month he again withdraws Birr 4,000 to spend steadily till the end of the month. In this scheme on an average he will be holding $\text{Birr } 4000/2 = 2000$ and will be investing remaining funds in saving deposits and earn interest on them. Thus, in this scheme he will be earning more interest income.

Thus, the greater the number of times an individual makes trips to the bank for withdrawing money, the greater the broker's fee he will incur. If he withdraws more cash, he will be avoiding some costs on account of brokerage fee.

Thus, individual faces a trade-off problem-, the greater the amount of pay cheque he withdraws in cash, less the cost on account of broker's fee but the greater the opportunity cost of forgoing interest income. The problem is therefore to determine an **optimum amount of money to hold**. Baumol has shown that optimal amount of money holding is determined by minimizing the cost of interest income forgone and broker's fee.

Let the size of the pay cheque (i.e. salary) be denoted by **Y**, the average amount of the cash he withdraws each time the individual goes to the bank by **C**, the number of times (**trips**) he goes to the bank to withdraw cash by **T**, broker's fee which he has to bear each time he makes a trip to the bank by **b**.

In the first scheme of money management when he gets his whole pay-cheque cashed on the first day of every month he incurs broker's fee only once since he makes only a single trip to the bank.

Thus T in our first case is equal to one $T = Y/C = 12000/12000 = 1$ because in this case $C = Y$. In the second, case, $T = 12000/6000 = 2$ and in the third case $T = 12000/4000 = 3$.

Interest income lost by holding money is the average amount of money holding multiplied by the interest rate. As seen above, average money held is one half of cash withdrawn each time (i.e., $C/2$).

Thus, interest income lost in the first case is $rC/2 = 5/100 \times 1200/2 = \text{Birr } 300$, in the second case interest lost $= r.C/2 = 5/100 \times 6000/2 = 150$ and in the third case it is $5/100 \times 4000/2 = 100$.

Thus the total cost incurred on broker's fee and interest income forgone is given by

$$\text{Total Cost} = bT + r.C/2$$

Where b stands for broker's fee

As seen above, $T = Y/C$

Therefore, Total Cost $= Yb/Cb + r.C/2$

Baumol has shown that average amount of cash withdrawal which minimises cost is given by

$$C = \sqrt{2bY/r}$$

This means that average amount of cash withdrawal which minimise cost is the square root of the two times broker's fee multiplied by the size of individual's income (Y) and divided by the interest rate. This is generally referred to as Square Root Rule.

For this rule, it follows that a higher broker's fee will raise the money holdings as it will discourage the individuals to make more trips to the bank. On the other hand, a higher interest rate will induce them to reduce their money holdings for transaction purposes as they will be induced to keep more funds in saving deposits to earn higher interest income. That is, at a higher rate of interest transactions demand for money holdings will decline.

Keynes thought that transactions demand for money was independent of rate of interest. According to him, transactions demand for money depends on the level of income. However, Baumol and Tobin have shown that transactions demand for money is sensitive to rate of interest.

As explained above, interest represents the opportunity cost of holding money instead of bonds, saving and fixed deposits. The higher the rate of interest, the greater the opportunity cost of holding money (i.e. the greater the interest income forgone for holding money for transactions).

Therefore, at a higher rate of interest people will try to economize the use of money and will demand less money for transactions. At a lower interest rate on bonds, saving and fixed deposits,

the opportunity cost of holding money will be less which will prompt people to hold more money for transactions.

Therefore, according to Baumol and Tobin, transactions demand curve for money slopes downward as shown in Fig. 19.4. At higher interest rates, bonds, savings and fixed deposits are more attractive relative to money holding for transactions.

Therefore, at higher interest rates people tend to hold less money for transaction purposes. On the other hand, when the rates of interest are low, opportunity cost of holding money will be less and, as a consequence, people will hold more money for transactions. Therefore, the curve of transaction demand for money slopes

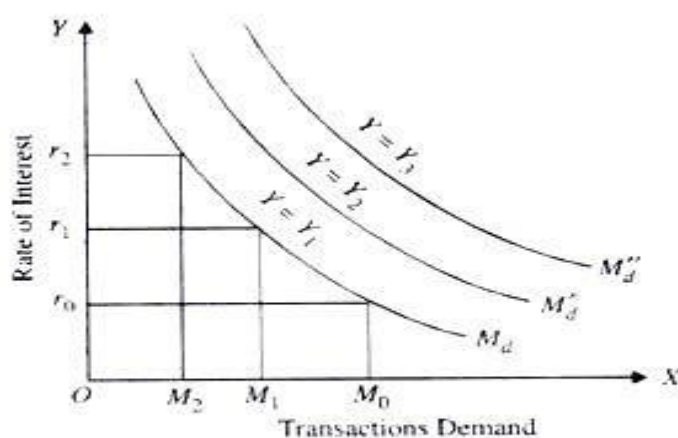


Fig. 19.4. Transactions Demand for Money : Baumol-Tobin Approach

downward.

It will be observed from the square root rule given above that transactions demand for money varies directly with the income (Y) of the individuals. Therefore, the higher the level of income, the greater the transactions demand for money at a given rate of interest.

In Fig. 19.4. The three transactions demand curves for money M_d , $M_{d'}$ and $M_{d''}$, for three different income levels, Y_1 , Y_2 , Y_3 are shown. It will be known from the square root rule that optimum money holding for transactions will increase less than proportionately to the increase in income. Thus, transactions demand for money, according to Baumol and Tobin, is function of both rate of interest and the level of income.

$$M_{td} = f(r, Y)$$

Where M_{td} stands for transactions demand for money, r for rate of interest and Y for the level of income.

3.5. Friedman's modern quantity theory of money

In 1956, Milton Friedman developed a theory of the demand for money in a famous article, —The Quantity Theory of Money: A Restatement. His analysis is closer to that of Keynes and the Cambridge economists than to Fisher's quantity theory of money though his article seems about quantity theory of demand. Though, Friedman, like his predecessors, pursued the question of why people choose to hold money, he did not however, deal with the specific motives for holding money, as Keynes did. He simply applied the general theory of portfolio choice to money. He said that the demand for money is affected by the same factors that affect the demand for any other asset.

Accordingly, the demand for any asset is affected by the relative return of its substitute assets in addition to the total wealth of the individual. More specifically, he claimed that the demand for money is affected by the resources available to the individual (wealth) and the expected returns of on other assets (which can serve as substitute for money) relative to the expected return on money.

□ He identified bonds, equities (stocks) and real goods as substitutes for money and hence their relative return will affect the demand for money. Like Keynes, he recognized that people want to hold a certain amount of real money balance, i.e., he put his demand function for money in real terms and included wealth and the relative returns of bond, stock and real goods as factors. His demand function is given below:

1. Wealth (W):

The major factor determining the demand for money is the wealth of the individual (W) In wealth Friedman includes not only non-human wealth such as bonds, shares, money which yield various rates of return but also human wealth or human capital. By human wealth Friedman means the value of an individual's present and future earnings.

Whereas non-human wealth can be easily converted into money, that is, can be made liquid. Such substitution of human wealth is not easily possible. Thus human wealth represents illiquid component of wealth and, therefore, the proportion of human wealth to the non-human wealth has been included in the demand for money function as an independent variable.

Individual's demand for money directly depends on his total wealth. Indeed, the total wealth of an individual represents an upper limit of holding money by an individual and is similar to the budget constraint of the consumer in the theory of demand.

The greater the wealth of an individual, the more money he will demand for transactions and other purposes. As a country, becomes richer, its demand for money for transaction and other purposes will increase.

Since as compared to non-human wealth, human wealth is much less liquid,

Friedman has argued that as the proportion of human wealth in the total wealth increases, there will be a greater demand for money to make up for the illiquidity of human wealth.

2. Rates of Interest or Return (r_m , r_b , r_e):

Friedman considers three rates of interest, namely, r_b , r_e and which determine the demand for money, r_m is the own rate of interest on money. Note that money kept in the form of currency and demand deposits does not earn any interest.

But money held as saving deposits and fixed deposits earns certain rates of interest and it is this rate of interest which is designated by r_m in the money demand function. Given the other rates of interest or return, the higher the own rate of interest, the greater the demand for money.

In deciding how large a part of his wealth to hold in the form of money the individual will compare the rate of interest on money with rates of interest (or return) on bonds and other assets. As mentioned earlier, the opportunity cost of holding money is the interest or return given up by not holding these other forms of assets.

As rates of return on bond (r_b) and equities (r_e) rise, the opportunity cost of holding money will increase which will reduce the demand for money holdings. Thus, the demand for money is negatively related to the rate of interest (or return) on bonds, equities and other such non-money assets.

Price Level (P):

Price level also determines the demand for money balances. A higher price level means people will require a larger nominal money balances in order to do the same amount of transactions, that is, to purchase the same amount of goods and services.

If income (Y) is used as proxy for wealth (W) which, as stated above, is the most important determinant of demand for money, then nominal income is given by $Y.P$ which becomes a crucial determinant of demand for money.

Here Y stands for real income (i.e. in terms of goods and services) and P for price level. As the price level goes up, the demand for money will rise and, on the other hand, if price level falls, the demand for money will decline. As a matter of fact, people adjust the nominal money balances (M) to achieve their desired level of real money balances (M/P).

The Expected Rate of Inflation ($\Delta P/P$):

If people expect a higher rate of inflation, they will reduce their demand for money holdings. This is because inflation reduces the value of their money balances in terms of its power to purchase goods and services.

If the rate of inflation exceeds the nominal rate of interest, there will be negative rate of return on money. Therefore, when people expect a higher rate of inflation they will tend to convert their money holdings into goods or other assets which are not affected by inflation.

On the other hand, if people expect a fall in the price level, their demand for money holdings will increase.

Institutional Factors (U):

Institutional factors such as mode of wage payments and bill payments also affect the demand for money. Several other factors which influence the overall economic environment affect the demand for money. For example, if recession or war is anticipated, the demand for money balances will increase.

Besides, instability in capital markets, which erodes the confidence of the people in making profits from investment in bonds and equity shares will also raise the demand for money. Even political instability in the country influences the demand for money. To account for these institutional factors Friedman includes the variable U in his demand for money function.

Simplifying Friedman's Demand for Money Function:

A major problem faced in using Friedman's demand for money function has been that due to the non-existence of reliable data about the value of wealth (W), it is difficult to estimate the demand for money. To overcome this difficulty Friedman suggested that since the present value of wealth

or $W = Y_p/r$ (where Y_p is the permanent income and r is the rate of interest on money.), permanent income Y_p can be used as a proxy variable for wealth.

Incorporating this in Friedman's demand for money function we have:

$$M_d = (Y_p, h, r_m, r_b, r_e, \Delta P/P, U)$$

If, we assume that no price change is anticipated and institutional factors such as h and U remain fixed in the short run and also all the three rates of interest return are clubbed into one, Friedman's demand for money function is simplified to

$$M_d = f(Y_p, r)$$

3.6. Empirical evidence on demand for money

Tobin's mean-variance analysis of money demand is just an application of the basic ideas in the theory of portfolio choice. Tobin assumes that the utility that people derive from their assets is positively related to the expected return on their portfolio of assets and is negatively related to the riskiness of this portfolio as represented by the variance (or standard deviation) of its returns.

Tobin's model then yields the same result as Keynes's analysis of the speculative demand for money: It is negatively related to the level of interest rates. This model, however, makes two important points that Keynes's model does not:

1. Individuals diversify their portfolios and hold money and bonds at the same time.
2. Even if the expected return on bonds is greater than the expected return on money, individuals will still hold money as a store of wealth because its return is more certain. Here we examine the empirical evidence on the two primary issues that distinguish the different theories of money demand and affect their conclusions about whether the quantity of money is the primary determinant of aggregate spending: Is the demand for money sensitive to changes in interest rates, and is the demand for money function stable over time?

The evidence on the interest sensitivity of the demand for money found by different researchers is remarkably consistent. Neither extreme case is supported by the data: The demand for money is sensitive to interest rates, but there is little evidence that a liquidity trap has ever existed. The evidence on the stability of the demand for money function is related to the evidence on the existence of a liquidity trap. Brunner and Meltzer's finding that a money demand function esti-

mated using data mostly from the 1930s predicted the demand for money well in the postwar period not only suggests that a liquidity trap did not exist in the 1930s, but also indicates that the money demand function has been stable over long periods.

Chapter Four

The Money supply Process

Introduction

In most countries today, a central bank or other monetary authority is charged with issuing domestic currency. That is an important charge because the supply of money greatly influences interest and inflation rates and, ultimately, aggregate output. If the central bank's monetary policy is good, if it creates just the right amount of money, the economy will hum, and interest and inflation rates will be low. If it creates too much money too quickly, prices will increase rapidly and wipe out people's savings until even the poorest people are nominal billionaires (as in Zimbabwe recently).

□ If it creates too little money too slowly, prices will fall, wiping out debtors and making it nearly impossible to earn profits in business (as in the Great Depression). But even less extreme errors can have serious negative consequences for the economy and hence your wallets, careers, and dreams. This chapter is a little involved, but it is worth thoroughly understanding the money supply process and money multipliers if you want to better understand how banks in particular and the financial system in general create money

□ Since deposits at banks are by far the largest component of the money supply, understanding how these deposits are created is the first step in understanding the money supply process. This chapter also provides an overview of how the banking system creates deposits.

4.1. Players in the Money Supply Process

The four players in the money supply process.

1. The central bank – the government agency that oversees the banking system and is responsible for the conduct of monetary policy. This bank takes different names in

different countries; the Federal reserve system in the United States and National Bank in Ethiopia, for instance.

2. Banks (depository institutions) – the financial intermediaries that accept deposits from individuals and institutions and make loans: commercial banks, savings and loan associations, mutual savings banks, and credit unions.
3. Depositors – individuals and institutions that hold deposits in banks.
4. Borrowers from banks – individuals and institutions that borrow from the depository institutions and institutions that issue bonds that are purchased by the depository institutions.

4.2. The central bank balance sheet and the monetary base

4.2.1. The central bank Balance sheet

Just as any other bank has a balance sheet that lists its assets and liabilities, so does the central bank. We examine each of the categories of assets and liabilities because changes in them have an important impact on the money supply.

Assets:

1. **Securities** – these are the central banks holdings of securities, which consist primarily of treasury securities. The total amount of securities is controlled by open market operations (the central bank's purchase and sale of these securities). Securities are by far the largest category of assets in the central banks' balance sheet.
2. **Discount loans**- these are loans the Central Bank (NBE) makes to banks, and the amount is affected by the Central bank's setting the discount rate, the interest rate the Central bank charges banks for these loans.

These first two Central bank assets are important because they earn interest. Because the liabilities of the Central bank do not pay interest, the Central bank makes millions of dollars every year. Its assets earn income and its liabilities cost nothing. Many central banks pay interest on reserves, but of course any interest paid is composed of cheap notes or, more likely, even cheaper accounting entries. Although it returns most of its earnings

to the federal government, the Central bank does spend some of it on —worthy causes, such as supporting economic research.

3. **Gold and SDR certificate accounts.** Special drawing rights (SDRS) are issued to governments by the International Monetary Fund (IMF) to settle international debts and have replaced gold in international financial transactions. When the treasury acquires gold or SDRs, it issues certificates to the Central bank that are claims on the gold or SDRs and is in turn credited with deposit balances at the Central bank. The gold and SDR accounts are made up of these certificates issued by the Treasury.
4. **Coin-** this is the smallest item in the balance sheet, and it consists of Treasury currency (mostly coins) held by the Central bank.
5. **Cash items in process of collection-** these arise from the Central bank's check-clearing process. When a check is given to the Central bank for clearing, the Central bank will present it to the bank on which it is written and will collect funds by deducting the amount of the check from the bank's deposits (reserves) with the Central bank. Before these funds are collected, the check is a cash item in process of collection and is a Central bank asset.
6. **Other Federal Reserve assets.** These include deposits and bonds denominated in foreign currencies as well as physical goods such as computers, office equipment, and buildings owned by the Central bank.

Liabilities

1. **Central bank notes (currency) outstanding:** The Central bank issues currency (those pieces of paper in your wallet). The Central bank notes outstanding are the amount of this currency that is in the hands of the public. (Currency held by depository institutions is also a liability of the Central bank but is counted as part of the reserves liability.)
2. **Reserves:** All banks have an account at the Central bank in which they hold deposits. Reserves consist of deposits at the Central bank plus currency that is physically held by banks (called vault cash because it is stored in bank vaults). Reserves are assets for the banks but liabilities for the Central bank because the banks can demand payment on them

at any time and the Central bank is required to satisfy its obligation by paying Central bank notes. As you will see, an increase in reserves leads to an increase in the level of deposits and hence in the money supply.

□ **Total reserves** can be divided in to two categories: reserves that the Central bank requires banks to hold (required reserves) and any additional reserves the banks choose to hold (excess reserves). For example, the Central bank might require that for every birr deposits at a depository institution, a certain fraction (say, 10 cents) must be held as reserves. This fraction (10 percent) is called the required reserve ratio.

3. **Treasury deposits:** The Treasury keeps deposits at the Central bank, against which it writes all its checks.
4. **Foreign and other deposits,** these include the deposits with the Central bank owned by foreign governments, foreign central banks, and international agencies.
5. **Deferred-availability cash items:** Like cash items in process of collection, these also arise from the Central bank's check-clearing process.
6. **Other liabilities and capital accounts:** this item includes all the remaining liabilities not included elsewhere on the balance sheet. For example, stock in the Central bank purchased by member banks is included here.

4.2.3 Control of the Monetary Base

The Central bank exercises control over the monetary base via the first two factors listed.: through its purchases or sales of government securities in the open market, called open market operations, and through its extension of discount loans to banks.

i. Open Market Operations (OMO)

The primary way in which the Central bank causes changes in the monetary base is through its open market operations. A purchase of bonds by the Central bank is called an **open market purchase**, and a sale of bonds by the Central bank is called an **open market sale**.

Open Market Purchase From A Bank: Suppose that the National Bank of Ethiopia purchases Birr 100 of bonds from a bank and pays for them with a Birr 100 check. The bank will either

deposit the check in its account with the National bank or cash it in for currency, which will be counted as vault cash. To understand what occurs as a result of this transaction, we look at T-accounts, which list only the changes that occur in balance sheet items starting from the initial balance sheet position. Either action means that the bank will find itself with Birr 100 more reserves and a reduction in its holdings of securities of Birr 100. The T-account for the banking system, then, is

Banking System		
Assets		Liabilities
Securities	-\$100	
Reserves		
+\$100		

The National bank meanwhile finds that its liabilities have increased by the additional Birr 100 of reserves, while its assets have increased by the Birr 100 of additional securities that it now holds. Its T-account is

National bank of Ethiopia		
Assets		Liabilities
Securities	+\$100	Reserves
		+\$100

The net result of this open market purchase is that reserves have increased by \$100, the amount of the open market purchase. Because there has been no change of currency in circulation, the monetary base has also risen by \$100.

Open Market Purchase from the Nonbank Public:

To understand what happens when there is an open market purchase from the nonbank public; we must look at two cases. First, let's assume that the person or corporation that sells the \$100 of bonds to the Central bank deposits the Central bank's check in the local bank. The nonbank public's T-account after this transaction is

Nonbank Public		
Assets		Liabilities
Securities	-\$100	
Checkable deposits		
+\$100		

When the bank receives the check, it credits the depositor's account with the \$100 and then deposits the check in its account with the Central bank, thereby adding to its reserves. The banking system's T-account becomes

Banking System		
Assets		Liabilities
Reserves	+\$100	Checkable deposits
		+\$100

The effect on the Central bank's balance sheet is that it has gained \$100 of securities in its assets column, while it has an increase of \$100 of reserves in its liabilities column:

Central bank

Assets		Liabilities	
Securities		Reserves	+\$100
+\$100			

As you can see in the above T-account, when the Central bank's check is deposited in a bank, the net result of the Central bank's open market purchase from the nonbank public is identical to the effect of its open market purchase from a bank: Reserves increase by the amount of the open market purchase, and the monetary base increases by the same amount.

If, however, the person or corporation selling the bonds to the Central bank cashes the Central bank's check either at a local bank or at a Central bank for currency, the effect on reserves is different. This seller will receive currency of \$100 while reducing holdings of securities by \$100. The bond seller's T-account will be

Nonbank Public			
Assets		Liabilities	
Securities	-\$100		
Currency	+\$100		

The Central bank now finds that it has exchanged \$100 of currency for \$100 of securities, so its T-account is

Central bank			
Assets		Liabilities	
Securities	+\$100	Currency in Circulation	+\$100

The net effect of the open market purchase in this case is that reserves are unchanged, while currency in circulation increases by the \$100 of the open market purchase. Thus the monetary base increases by the \$100 amount of the open market

purchase, while reserves do not. This contrasts with the case in which the seller of the bonds deposits the Central bank's check in a bank; in that case, reserves increase by \$100, and so does the monetary base.

The analysis reveals that the effect of an open market purchase on reserves depends on whether the seller of the bonds keeps the proceeds from the sale in currency or in deposits.

□ If the proceeds are kept in Currency, the open market purchase has no effect on reserves; if the proceeds are kept as deposits, reserves increase by the amount of the open market purchase.

✎ The effect of an open market purchase on the monetary base, however, is always the same (the monetary base increases by the amount of the purchase) whether the seller of the bonds keeps the proceeds in deposits or in currency. The impact of an open market purchase on reserves is much more uncertain than its impact on the monetary base.

OPEN MARKET SALE (OMS): If the Central bank sells \$100 of bonds to a bank or the nonbank public; the monetary base will decline by \$100. For example, if the Central bank sells the bonds to an individual who pays for them with currency, the buyer exchanges \$100 of currency for \$100 of bonds, and the resulting T-account is

Nonbank Public		
	Assets	Liabilities
Securities	+\$100	
Currency	-\$100	

□ The Central bank, for its part, has reduced its holdings of securities by \$100 and has also lowered its monetary liability by accepting the currency as payment for its bonds, thereby reducing the amount of currency in circulation by \$100:

Central bank		
	Assets	Liabilities
Securities	-\$100	Currency in Circulation
		-\$100

□ The effect of the open market sale of \$100 of bonds is to reduce the monetary base by an equal amount, although reserves remain 'unchanged. Manipulations of T-accounts in

cases in which the buyer of the bonds is a bank or the buyer pays for the bonds with a check written on a checkable deposit account at a local bank lead to the same \$100 reduction in the monetary base, although the reduction occurs because the level of reserves has fallen by \$100.

□ The following conclusion can be drawn from our analysis of open market purchases and sales: The effect of OMO on the monetary base is much more certain than the effect on reserves. Therefore, the central bank can control the monetary base with OMO more effectively than it can control reserves.

 Open market operations can also be done in other assets besides government bonds have the same effects on the monetary base we have described here. One example of this is a foreign exchange intervention by the Central bank.

Shifts from Deposits into currency

 Even if the Central bank does not conduct open market operations, a shift from deposits to currency will affect the reserves in the banking system. However, such a shift will have no effect on the monetary base, another reason why the Central bank has more control over the monetary base than over reserves.

□ Let's suppose that W/ro Roman (who opened a \$100 checking account at the Commercial bank of Ethiopia) decides that tellers are so abusive in all banks that she closes her account by withdrawing the \$100 balance in cash and vows never to deposit it in a bank again. The effect on the T-account of the nonbank public is

Nonbank Public		
Assets		Liabilities
Checkable deposits	-\$100	
Currency	+\$100	

The banking system loses \$100 of deposits and hence \$100 of reserves:

Banking System		
Assets		Liabilities

Reserves	\$100	Checkable deposits	-\$100
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For the Central bank, W/ro Roman's action means that there is \$100 of additional Currency circulating in the hands of the public, while reserves in the banking system have fallen by \$100. The Central bank's T-account is

Central Bank			
Assets		Liabilities	
		Currency in Circulation	+\$100
		Reserves	-\$100

□ The net effect on the monetary liabilities of the Central bank is a wash; the monetary base is unaffected by W/ro Roman's disgust at the banking system. But reserves are affected. Random fluctuations of reserves can occur as a result of random shifts into currency and out of deposits, and vice versa. The same is not true for the monetary base, making it a more stable variable.

ii. Discount Loans

□ In this chapter so far we have seen changes in the monetary base solely as a result of open market operations. However, the monetary base is also affected when the Central bank makes a discount loan to a bank. When the Central bank made a \$100 discount loan to the Commercial Bank of Ethiopia, the bank was credited with \$100 of reserves from the proceeds of the loan. The effects on the balance sheet of the banking system and the Central bank are illustrated by the following T-accounts:

Banking System				Central Bank			
Asset		Liabilities		Asset		Liabilities	
Reserve	+\$100	Discount Loans	+\$100	Discount	+\$100	Reserve	+\$100
				Loans			

□ The monetary liabilities of the Central bank have now increased by \$100 and the monetary base, too, has increased by this amount. However, if a bank pays off a loan from the Central bank, thereby reducing its borrowings from the Central bank by \$100, the T-accounts of the banking system and the Central bank are as follows:

Banking system				Central Bank			
Asset		Liabilities		Asset		Liabilities	
Reserve	-\$100	Discount	-\$100	Discount	-\$100	Reserve	-\$100
		Loans		Loans			

The net effect on the monetary liabilities of the Central Bank, and hence on the monetary base, is then a reduction of \$100. We see that the monetary base changes one-for-one with the change in the borrowings from the Central bank.

4.2.4 Overview of the Central Bank's Ability to Control the Monetary Base

The factor that most affects the monetary base is the Central bank's holdings of securities, which are completely controlled by the Central bank through its open market operation. Factors not controlled by the Central bank (for example, float and Treasury deposits with the Central bank) undergo substantial short – run variations and can be important sources of fluctuation in the monetary base over time periods as short as a week. However, these fluctuations are usually quite predictable and so can be offset through open market operations. Although float and Treasury deposits with the Fed undergo substantial short run fluctuations, which complicate control of the monetary base, they do not prevent the Central bank from accurately controlling it.

4.3 Multiple Deposit Creation: A Simple Model

4.3.1 Deposit Creation: the Single Bank

Suppose that the \$100 open market purchase described earlier was conducted with the Commercial Bank of Ethiopia. After the Central bank has bought the \$100 bond from the Commercial Bank of Ethiopia, the bank finds that it has an increase in reserves of \$100. To analyze what the bank will do with these additional reserves, assume that the bank does not want

to hold excess reserves because **it earns no interest on them**. We begin the analysis with the following T-account:

Commercial Bank of Ethiopia		
	Assets	Liabilities
Securities	-\$100	
Reserves	+\$100	

 Because the bank has no increase in its checkable deposits, required reserves remain the same, and the bank finds that its additional \$100 of reserves means that its excess reserves have increased by \$100. Let's say that the bank decides to make a loan equal in amount to the \$100 increase in excess reserves. When the bank **makes the loan, it sets up a checking account for** the borrower and puts the proceeds of the loan into this account. In this way the bank alters its balance sheet by increasing its liabilities with \$100 of checkable deposits and at the same time increasing its assets with the \$100 loan. The resulting T-account looks like this:

Commercial Bank of Ethiopia		
	Assets	Liabilities
Securities	-\$100	
Reserves	+\$100	
Loans	+\$100	

□ The bank has created checkable deposits by its act of lending. Because checkable deposits are part of the money supply, the bank's act of lending has in fact created money.

□ In its current balance sheet position, the Commercial Bank of Ethiopia still has excess reserves and so might want to make additional loans. However, these reserves will not stay at the bank for very long. The borrower took out a loan not to leave \$100 idle at the Commercial Bank of Ethiopia but to purchase goods and services from other individuals and corporations. When the borrower makes these purchases by writing checks, they will be deposited at other banks, and the \$100 of reserves will leave the Commercial Bank of Ethiopia. A bank cannot safely make loans for an amount greater than the excess reserves it has before it makes the loan. The final T-account of the Commercial Bank of Ethiopia is

Commercial Bank of Ethiopia

Assets		Liabilities
Securities	-\$100	
Loans	+\$100	

□ The increase in reserves of \$100 has been converted into additional loans of \$100 at the Commercial Bank of Ethiopia, plus an additional \$100 of deposits that have made their way to other banks. (All the checks written on accounts at the Commercial Bank of Ethiopia are deposited in banks rather than converted into cash because we are assuming that the public does not want to hold any additional currency.) Now let's see what happens to these deposits at the other banks.

4.3.2 Deposit Creation: The Banking System

To simplify the analysis, let us assume that the \$100 of deposits created by Commercial Bank of Ethiopia's loan is deposited at Wegagen Bank and that this bank and all other banks hold no excess reserves. Wegagen's T-account becomes:

Wegagen Bank			
Assets		Liabilities	
Reserves	+\$100	Checkable deposits	+\$100

□ If the required reserve ratio is 10 percent, this bank will now find itself with a \$10 increase in required reserves, leaving it \$90 of excess reserves. Because Wegagen Bank (like the Commercial Bank of Ethiopia) does not want to hold on to excess reserves, it will make loans for the entire amount. Its loans and checkable deposits will then increase by \$90, but when the borrower spends the \$90 of checkable deposits, they and the reserves at Bank A will fall back down by this same amount. The net result is that Wegagen Bank's T-account will look like this:

Wegagen Bank			
Assets		Liabilities	
Reserves	+\$10	Checkable deposits	+\$100
Loans	+\$90		

If the money spent by the borrower to whom Wegagen Bank lent the \$90 is deposited in another bank, such as Awash International Bank, the T-account for Awash International Bank will be

Awash International Bank

Assets		Liabilities	
Reserves	+\$90	Checkable deposits	+\$90

☞ The checkable deposits in the banking system have increased by another \$90, for a total increase of \$190 (\$100 at Wegagen Bank plus \$90 at Awash International Bank). In fact, the distinction between Wegagen Bank and Awash International Bank is not necessary to obtain the same result on the overall expansion of deposits. If the borrower from Wegagen Bank writes checks to someone who deposits them at Wegagen Bank, the same change in deposits would occur. The T-accounts for Awash International Bank would just apply to Wegagen Bank, and its checkable deposits would increase by the total amount of \$190.

☞ Awash International Bank will want to modify its balance sheet further. It must keep 10 percent of \$90 (\$9) as required reserves and has 90 percent of \$90 (\$81) in excess reserves and so can make loans of this amount. Awash International Bank will make an \$81 loan to a borrower, who spends the proceeds from the loan. Awash International Bank T-account will be

Awash International Bank			
Assets		Liabilities	
Loans	+\$81	Reserves	+\$9
		Checkable deposits	+\$90

The \$81 spent by the borrower from Awash International Bank will be deposited in another bank (Dashen Bank). Consequently, from the initial \$100 increase of reserves in the banking system, the total increase of checkable deposits in the system so far is \$271 (= \$100 + \$90 + \$81). □

Following the same reasoning, if all banks make loans for the full amount of their excess reserves, further increments in checkable deposits will continue (at Dashen Bank, Abyssinia Bank, United Bank, and so on), as depicted in Table 4.1. Therefore, the **total increase in deposits from the initial \$100 increase in reserves will be \$1000: The increase is tenfold, the reciprocal of the 0.10 reserve requirement.**

□ If the banks choose to invest their excess reserves in securities, the result is the same. If

Wegagen Bank had taken its excess reserves and purchased securities instead of making loans, its T-account would have looked like this:

Wegagen Bank			
Assets		Liabilities	
Reserves	+\$10	Checkable deposits	+\$100
Securities	+\$90		

□ When the bank buys \$90 of securities, it writes a \$90 check to the seller of the securities, who in turn deposits the \$90 at a bank such as Awash International Bank. Awash International Bank's checkable deposits rise by \$90 and the deposit expansion process is the same as before. Whether a bank chooses to use its excess reserves to make loans or to purchase securities, the effect on deposit expansion is the same.

Table4. 1 Creation of Deposits (assuming 10 percent reserve requirement and a \$100 increase in reserves)

Bank

	Deposits (\$)	Loans (\$)	Reserves (\$)
Commercial Bank of Ethiopia	0.00	100.00	0.00
Wegagen Bank	100.00	90.00	10.00
Awash International Bank	90.00	81.00	9.00
Dashen	81.00	72.90	8.10
Abyssinia Bank	72.90	65.61	7.29
United Bank	65.61	59.05	6.56
Nib International Ban	59.05	53.14	5.91
.....			
Total for all bank	1000.00	1000.00	100.00

You can now see the difference in deposit creation for the single bank versus the banking system as a whole. Because a single bank can create deposits, equal only to the amount of its excess reserves, it cannot by itself generate multiple deposit expansion. A single bank cannot make loans greater in amount than its excess reserves because the bank will lose these reserves as the deposits created by the loan find their way to other banks. However, the banking system as a whole can generate a multiple expansion of deposits because when a bank loses its excess reserves, these

reserves do not leave the banking System even though they are lost to the individual bank. So as each bank makes a loan and creates deposits, the reserves find their way to another bank, which uses them to make additional loans and create additional deposits. As you have seen, this process continues until the initial increase in reserves results in a multiple increase in deposits.

The multiple increases in deposits generated from an increase in the banking system's reserves is called the **simple deposit multiplier**. In our example with a 10 percent required reserve ratio, the simple deposit multiplier is 10. More generally, the simple deposit multiplier equals the reciprocal of the required reserve ratio, expressed as a fraction ($10 = 1/0.10$), so the formula for the multiple expansions of deposits can be written as

$$\Delta D = \frac{1}{r_D} \times \Delta R \dots\dots\dots (2)$$

Where: ΔD = change in total checkable deposits in the banking system

r_D = required reserve ratio (0.10 in the example)

ΔR = change in reserves for the banking system (\$100 in the example)

4.4. Monetary Supply Model and Money Multiplier

In our model of multiple deposit creation above, we ignored the effects on deposit creation of changes in the public's holdings of currency and banks' holdings of excess reserves. Now we incorporate these changes into our model of the money supply process by assuming that the desired level of currency C and excess reserves ER grows proportionally with checkable deposits D ; in other words, we assume that the ratios of these items to checkable deposits are constants in equilibrium:

Assume the desired level of currency C and excess reserves ER
grows proportionally with checkable deposits D

Then

$c = \{C / D\}$ = currency ratio

$e = \{ER / D\}$ = excess reserves ratio

The total amount of reserves (R) equals the sum of required reserves (RR) and excess reserves (ER).

$$R = RR + ER$$

The total amount of required reserves equals the required reserve ratio times the amount of checkable deposits

$$RR = r \times D$$

Substituting for RR in the first equation

$$R = (r \times D) + ER$$

The Fed sets r to less than 1

The monetary base MB equals currency (C) plus reserves (R)

$$MB = R + C = (r \times D) + ER + C$$

Reveals the amount of the monetary base needed to support the existing amounts of checkable deposits, currency, and excess reserves.

An increase in the monetary base that goes into currency is not multiplied, whereas an increase that goes into supporting deposits is multiplied.

An additional dollar of MB that goes into excess reserves ER does not support any additional deposits or currency

$$c = \{C / D\} \Rightarrow C = c \times D \text{ and}$$

$$e = \{ER / D\} \Rightarrow ER = e \times D$$

Substituting in the previous equation

$$MB = (r \times D) + (e \times D) + (c \times D) = (r + e + c) \times D$$

Divide both sides by the term in parentheses

$$D = \frac{1}{r + e + c} \times MB$$

$$M = D + C \text{ and } C = c \times D$$

$$M = D + (c \times D) = (1 + c) \times D$$

Substituting again

$$M = \frac{1 + c}{r + e + c} \times MB$$

The money multiplier is then

$$m = \frac{1 + c}{r + e + c}$$

Intuition Behind the Money Multiplier

r = required reserve ratio = 0.10

C = currency in circulation = \$400B

D = checkable deposits = \$800B

ER = excess reserves = \$0.8B

M = money supply ($M1$) = $C + D = \$1,200B$

$$c = \frac{\$400B}{\$800B} = 0.5$$

$$e = \frac{\$0.8B}{\$800B} = 0.001$$

$$m = \frac{1 + 0.5}{0.1 + 0.001 + 0.5} = \frac{1.5}{0.601} = 2.5$$

This is less than the simple deposit multiplier

Although there is multiple expansion of deposits,

there is no such expansion for currency

4.5 Factors that Determine the Money Multiplier

To develop our intuition of the money multiplier even further, let us look at how this multiplier changes in response to changes in the variables in our model: $\{C/D\}$, $\{ER/D\}$, and r_D . The "game" we are playing is a familiar one in economics: We ask what happens when one of these variables changes, leaving all other variables the same (*ceteris paribus*).

Changes in the Required Reserve Ratio r_D

If the required reserve ratio on checkable deposits increases while all the other variables stay the same, the same level of reserves cannot support as large an amount of checkable deposits; more reserves are needed because required reserves for these checkable deposits have risen. The resulting deficiency in reserves then means that banks must contract their loans, causing a decline in deposits and hence in the money supply.

✎ The reduced money supply relative to the level of MB, which has remained unchanged, indicates that the money multiplier has declined as well. Another way to see this is to realize that when r_D is higher, less multiple expansions of checkable deposits occur. With less multiple deposit expansion, the money multiplier must fall.

We can verify that the foregoing analysis is correct by seeing what happens to the value of the money multiplier in our numerical example when r_D increases from 10 percent to 15 percent

(leaving all the other variables unchanged). The money multiplier becomes

$$m = \frac{1+0.5}{0.15+0.001+0.5} = \frac{1.5}{0.651} = 2.3 \text{ which, as we would expect, is less than 2.5.}$$

The analysis just conducted can also be applied to the case in which the required reserve ratio falls. In this case, there will be more multiple expansion for checkable deposits because the same level of reserves can now support more checkable deposits, and the money multiplier will rise. For example, if r_D falls from 10 percent to 5 percent, plugging this value into our money multiplier formula (leaving all the other variables unchanged) yields a money multiplier of

$$m = \frac{1+0.5}{0.05+0.001+0.5} = \frac{1.5}{0.551} = 2.72 \text{ which is above the initial value of 2.5.}$$

We can now state the following result: *The money multiplier and the money supply are negatively related to the required reserve ratio r_D .*

Changes in the Currency Ratio {C/D}

Next, what happens to the money multiplier when depositor behavior causes {C/D} to increase with all other variables unchanged? An increase in {C/D} means that depositors are converting some of their checkable deposits into currency. As shown before; checkable deposits undergo multiple expansions while currency does not. Hence when checkable deposits are being converted into currency, there is a switch from a component of the money supply that undergoes, multiple expansion to one that does not. The overall level of multiple expansion declines, and so must the multiplier.

This reasoning is confirmed by our numerical example, where {C/D} rises from 0.50 to 0.75. The money multiplier then falls from 2.5 to

$$m = \frac{1+0.75}{0.1+0.001+0.75} = \frac{1.75}{0.851} = 2.06$$

We have now demonstrated another result: *The money multiplier and the money supply are negatively related to the currency ratio {C/D}.*

Changes in the Excess Reserves Ratio {ER/D}

When banks increase their holdings of excess reserves relative to checkable deposits, the banking system in effect has fewer reserves to support checkable deposits. This means that given the same level of MB banks will contract their loans, causing a

decline in the level of checkable deposits and a decline in the money supply, and the money multiplier will fall.

This reasoning is supported in our numerical example when $\{ER/D\}$ rises from 0.001 to 0.005. The money multiplier declines from 2.5 to

$$m = \frac{1 + 0.5}{0.1 + 0.005 + 0.5} = \frac{1.5}{0.605} = 2.48$$

Note that although the excess reserves ratio has risen, there has been only a small decline in the money multiplier. This decline is small because in recent years the $\{ER/D\}$ ratio has been extremely small, so changes in it have only a small impact on the money multiplier. However, there have been times, particularly during the Great Depression, when this ratio was far higher, and its movements had a substantial effect on the money supply and the money multiplier. Thus our final result is still an important one: *The money multiplier and the money supply are negatively related to the excess reserves ratio $\{ER/D\}$.*

Chapter Five: Central banking and the conduct of monetary policy

5.1 Central Banking

Central banks are relatively new inventions. An American President (Andrew Jackson) even cancelled his country's central bank in the nineteenth century because he did not think that it was very important. But things have changed since. Central banks today are the most important feature of the financial systems of most countries of the world.

Central banks are bizarre hybrids. Some of their functions are identical to the functions of regular, commercial banks. Other functions are unique to the central bank. On certain functions it has an absolute legal monopoly.

A nation's principal monetary authority, such as the Federal Reserve Bank (USA) and National Bank of Ethiopia, which regulates the money supply and credit, issues currency, and manages the rate of exchange.

It is the entity responsible for overseeing the monetary system for a nation (or group of nations). Central banks have a wide range of responsibilities - from overseeing monetary policy to implementing specific goals such as currency stability, low inflation and full employment. Central banks also generally issue currency, function as the bank of the government, regulate the credit system, oversee commercial banks, manage exchange reserves and act as a lender of last resort.

i. Activities and responsibilities of Central Banks

Functions of a central bank (not all functions carried out by all banks):

- monopoly on the issue of banknotes
- the Government's banker
- the bankers' bank ("Lender of Last Resort")
- manages the country's foreign exchange and gold reserves and the Government's stock register;
- regulation and supervision of the banking industry:
- setting the official interest rate - used to manage both inflation and the country's exchange rate - and ensuring that this rate takes effect via a variety of policy mechanisms

ii. Policy instruments of the Central Bank

Monetary policy instruments may be divided in to two main types.

i. **Direct controls**

Direct controls are typically directives given by the central bank to control the quantity or price (interest rate) of money deposited with commercial banks and credit provided by them. Ceilings on the growth of bank lending or deposits are examples of quantity controls. Maximum bank lending or deposit rates are examples of interest rate controls.

ii. **Indirect instruments**

An important reason for financial liberalization is to develop a system which promotes an efficient allocation of savings and credit in the economy. In the monetary area, financial liberalization involves a movement away from direct monetary controls towards indirect ones. The latter operate by the central bank controlling the price or volume of the supply of its own liabilities - reserve money - which in turn may affect interest rates more widely and the quantity of money and credit in the whole banking system. Indirect instruments used in monetary operations are often divided in to three:

1. Open Market Operations (OMO)

To enable open market operations, a central bank must hold foreign exchange reserves (usually in the form of government bonds) and official gold reserves. It will often have some influence over any official or mandated exchange rates: Some exchange rates are managed, some are market based (free float) and many are somewhere in between "managed float" or "dirty float". Through open market operations, a central bank influences the money supply in an economy directly. Each time it buys securities, exchanging money for the security, it raises the money supply. Conversely, selling of securities lowers the money supply. Buying of securities thus amounts to printing new money while lowering supply of the specific security.

The main open market operations are:

- Temporary lending of money for collateral securities ("Reverse Operations"). These operations are carried out on a regular basis, where fixed maturity loans are auctioned off.
- Buying or selling securities ("Direct Operations") on ad-hoc basis.
- Foreign exchange operations such as forex swaps.

2. Reserve Requirements

Changes in reserve requirements affect the money supply by causing the money supply multiplier to change. A rise in reserve requirements reduces the amount of deposits that can be supported by a given level of the monetary base and will lead to a contraction of the money supply. Conversely,

a decline in reserve requirements leads to an expansion of the money supply because more multiple deposit creation can take place.

2. Discount Policy

Discount policy, which primarily involves changes in the discount rate, affects the money by affecting the volume of discount loans and the monetary base. A rise in discount loans adds to the monetary base and expands the money supply; a fall in discount loans reduces the monetary base and shrinks the money supply.

iii. Central Bank Independence

Why is central bank independence an important issue?

The brief answer to this question is that price stability is generally considered a good thing, and that an independent central bank can help to achieve it. I do not need to dwell on the desirability of price stability. Economies work better if investment and wage decisions are not confused and thwarted by high inflation.

Some people see price stability as an anchor not only for the economy, but also for society at large. Price stability, however, is *not* the natural order of things in a modern economy.

Apart from higher oil prices and other shocks_, there are two particular threats which bear upon the issue of central bank independence:

- The tendency for policy makers and politicians to push the economy to run faster and further than its capacity limits allow; and
- The temptation that governments have to incur budget deficits and fund these by borrowings from the central bank.

Let us examine these two sources of inflationary pressure more closely. It is understandable that policy makers and politicians (in part in response to public pressure) should wish to squeeze as much growth as possible out of the economy – to run it a bit faster and a bit further than its capacity limits allow. In the jargon, there is a tendency to exploit the short-run trade-off between output and inflation. Even economies which have few spare resources can grow at rates above their long-term capacity limits for a time. Although it may take some time to show up, higher inflation will be the inevitable result.

Cynics blame this inflation bias on the political process, claiming that politicians have short time horizons, stretching only as far as the next election.

In principle, the answer to the first threat is relatively straightforward – give central banks a charter which includes a strong commitment to price stability, and the freedom to pursue it. This does involve the government in setting the goals, but that is the way it should be: central banks cannot expect to determine the goals they should pursue, but they should have adequate scope to pursue the goals that have been set. In the jargon, they should not have *goal independence* but they should have **instrument independence**.

To list a couple of points:

Legal independence, which is what the empirical studies seek to measure, can be very different from practical independence.

In Australia, the legislation provides that in the event of a dispute over monetary policy, the government can override the Reserve Bank by tabling its objections before both houses of parliament. While this is the legal position (which scores a negative on the index of independence), such a situation has never arisen in practice; and Indexes of legal independence cannot capture the role of personalities (such as the Governor and the Treasurer/Finance Minister) who fill in the operational gaps in the relevant legislation. Nor can they capture the changing policy-making environment.

In practice, it seems natural that a central bank will always consider the effects of its actions on the financial system as a whole. Moreover, if an individual bank should find itself in distress it will almost certainly require remedial action which is different in kind from that which could be afforded through deferment of monetary policy action. Tensions will sometimes arise whether the supervisory agency is within the central bank or external to it; in my mind, there is something to be said for internalizing these tensions.

5.3 Lags in Monetary Economics

Monetary indicators indicate the direction and strength of monetary policy in a particular period being closely related to the instruments of monetary policy. Indicators are not directly controllable by the authorities for two reasons.

Firstly, there may be uncertainty over the exact links between instruments and indicators. The actual structure of the economy may be unknown so that the authorities have no clear idea how an instrument and then indicator affects the goal variables. Within a well-established structure

itself, uncertainties may also be introduced because of the occurrence of stochastic variations in the basic functions. For example the demand for money function may shift due to an exogenous disturbance that may affect the result of any change in a particular monetary policy instrument.

External influences are introduced to the system in the following manner.

An illustration can be made on the above scheme: suppose that authorities undertake a selling operation in the market of government securities to reduce money supply (an intermediate target).

□ Despite this, the money supply increases. This unexpected reaction of the money supply can be due to a number of exogenous factors which may include the decision of the banking system to reduce its reserve ratio in response perhaps to an increase in the demand for credit. Alternatively an inflow of capital from abroad may inflate high powered money more than the open market operation has reduced it.

- ❖ In both cases money supply increased due to exogenous factors' influence on the indicator/ target.
- ❖ As the indicators are closer to the policy actions themselves, they will be less susceptible to exogenous factors than are the target variables. Secondly, problems arise due to the existence of lags in the implementation and effects of monetary policy actions. Though, it is the lagged response of goal variables to a policy change that necessitates the use of indicators and targets, at times they would be excessive and create inconvenience to the use of indicators and targets themselves.
- ❖ These lags are of two types: inside lags (implementation lags) and outside lags.
 1. **The inside lag:** - this is the delay that ensues between the need for a policy change and the policy adjustment actually being made. Inside lag in itself may be divided into a recognition lag- the time period between the need arising and its being recognized- and administrative lag- the time period between recognition and the policy change being made. The latter will be fairly short for monetary policy, as actions can soon be taken to alter the scale of monetary policy instruments.
 2. **The outside lag:** - this is the time elapsing between the policy instrument being adjusted and the effect being felt on the goal variables. This consists of the intermediate lag, the time period between the adjustment of the policy instrument and its effect on the major target variable; and the reaction lag, the time between the change in the target variable and the effect on the goal variable itself. The whole lag structure demonstrates the extent of the policy makers' problems with the indicators.

5.4 Conduct of Monetary Policy: Goals and Targets

□ Now that we understand the tools central banks like the Federal Reserve (USA) and National Bank (Ethiopia) use to conduct monetary policy, we can proceed to see how monetary policy is actually conducted by central banks. Understanding the conduct of monetary policy is important because it not only affects the money supply and interest rates but also has a major influence on the level of economic activity and hence on our wellbeing.

To explore this subject, we look at the goals that the central banks establish for monetary policy and their strategies for attaining them.

i. Goals of Monetary Policy

Six basic goals are continually mentioned by personnel at the Federal Reserve and other central banks when they discuss the objectives of monetary policy: (1) high employment, (2) economic growth, (3) price stability, (4) interest-rate stability, (5) stability of financial markets, and (6) stability in foreign exchange markets.

High Employment

Promoting high employment that is consistent with a stable price level is the main macroeconomic objective of Countries. High employment is a worthy goal for two main reasons: (1) the alternative situation, high unemployment, causes much human misery, with families suffering financial distress, loss of personal self-respect, and increase in crime (though this last conclusion is highly controversial) and (2) when unemployment is high, the economy has not only idle workers but also idle resource (closed factories and unused equipment), resulting in a loss of output (lower GDP).

Although it is clear that high employment is desirable, how high should it be? At What point can we say that the economy is at full employment? At first, it might seem that full employment is the point at which no worker is out of a job, that is, when unemployment is zero. But this definition ignores the fact that some unemployment, called frictional unemployment, which involves searches by workers and firms to find suitable matchups, is beneficial to the economy.

□ For example, a worker who decides to look for a better job might be unemployed for a while during the job search. Workers often decide to leave work temporarily to pursue other activities (raising a family, travel, returning to school), and when they decide to reenter the job market, it

may take some time for them to find the right job. Another reason that unemployment is not zero when the economy is at full employment is due to what is called structural unemployment, a mismatch between job requirements and the skills or availability of local workers. Clearly, this kind of unemployment is undesirable. Nonetheless, it is something that monetary policy can do little about.

□ The goal for high employment should therefore not seek an unemployment level of zero but rather a level above zero consistent with full employment at which the demand for labor equals the supply of labor. This level is called the **natural rate of unemployment**.

Economic Growth

The goal of steady economic growth is closely related to the high-employment goal because businesses are more likely to invest in capital equipment to increase Productivity and economic growth when unemployment is low. Conversely, if unemployment is high and factories are idle, it does not pay for a firm to invest in additional plants and equipment.

Price Stability

Price stability is desirable because a rising price level (inflation) creates uncertainty in the economy, and that may hamper economic growth. For example, the information conveyed by the prices of goods and services is harder to interpret when the overall level of prices is changing, which complicates decision making for consumers, businesses, and government.

Inflation also makes it hard to plan for the future, For example, it is more difficult to decide how much funds should be put aside to provide for a child's college education in an inflationary environment. Further, inflation may strain a country's social fabric: Conflict may result because each group in the society may compete with other groups to make sure that its income keeps up with the rising level of prices.

Interest-Rate Stability

Interest-rate stability is desirable because fluctuations in interest rates can create uncertainty in the economy and make it harder to plan for the future. Fluctuations in interest rate that affect consumers' willingness to buy house, for example, make it more difficult for consumers to decide when to purchase a house and for construction firms to plan how many houses to build. A central bank may also want to reduce upward movements in interest rates for the reasons we discussed at

the beginning of this chapter. Upward movements in interest rates generate hostility toward central banks like the Fed and lead to demands that their power be curtailed.

Stability of Financial Markets

Financial crises can interfere with the ability of financial markets to channel funds to people with productive investment opportunities, thereby leading to a sharp contraction in economic activity. The promotion of a more stable financial system in which financial crises are avoided is thus an important goal for a central bank.

The stability of financial markets is also fostered by interest-rate stability because fluctuations in interest rates create great uncertainty for financial institutions. An increase in interest rates produces large capital losses on long-term bonds and mortgages; losses that can cause the failure of the financial institutions holding them.

□ In recent years, more pronounced interest-rate fluctuations have been a particularly severe problem for savings and loan associations and mutual savings banks, many of which got into serious financial trouble in the 1980s and early 1990s.

Stability in Foreign Exchange Markets

With the increasing importance of international trade to a country's economy, the value of its currency relative to other currencies (such as dollar) has become a major consideration for the Central Banks. For example a rise in the value of the Birr makes Ethiopian industries less competitive with those abroad, and declines in the value of the Birr stimulate inflation in Ethiopia. In addition, preventing large changes in the value of the Birr makes it easier for firms and individuals purchasing or selling goods abroad to plan ahead.

Stabilizing extreme movements in the value of the dollar in foreign exchange markets is thus viewed as a worthy goal of monetary policy.

In other countries, which are even more dependent on foreign trade, stability in foreign exchange markets takes on even greater importance.

Conflict among Goals

Although many of the goals mentioned are consistent with each other-high employment with economic growth, interest-rate stability with financial market stability-this is not always the case. The goal of price stability often conflicts with the goals of interest-rate stability and high employment in the short run (but probably not in the long run).

□ For example, when the economy is expanding and unemployment is falling, both inflation and interest rates may start to rise. If the central bank tries to prevent a rise in interest rates, this may cause the economy to overheat and stimulate inflation. But if a central bank raises interest rates to prevent inflation, in the short run unemployment may rise. The conflict among goals may thus present central banks with some hard choices. We return to the issue of how central banks should choose conflicting goals in later chapters when we examine how monetary policy affects the economy.

ii. Central Bank Strategy: Use of Targets

The central bank's problem is that it wishes to achieve certain goals, such as price stability with high employment, but it does not directly influence the goals. It has a set of tools to employ (open market operations, changes in the discount rate, and changes in reserve requirements) that can affect the goals indirectly after a period of time (typically more than a year). If the central bank waits to see what the price level and employment will be one year later, it will be too late to make any corrections to its policy-mistakes will be irreversible.

All central banks consequently pursue a different strategy for conducting monetary policy by aiming at variable that lie between its tools and the achievement of its goals. The strategy is as follows:

After deciding on its goals for employment and the price level, the central bank chooses a set of variables to aim for, called **intermediate targets**, such as the monetary aggregates (M1, M2, or M3) or interest rates (short- or long-term), which have direct effect on employment and the price level. However, even these intermediate targets are not directly affected by the central bank's policy tools.

Therefore, it chooses another set of variables to aim for, called **operating targets**, or alternatively called instruments, such as reserve aggregates (reserves, non-borrowed reserves, monetary base, or non-borrowed base) or interest rates (federal funds rate or Treasury bill rate), which are more responsive to its policy tools. (Recall that non-borrowed reserves are total, reserves minus borrowed reserves; which are the amount of discount loans; the non-borrowed base is the monetary base minus borrowed reserves; and the federal funds rate is the interest rate on funds loaned overnight between banks.)

The central bank pursues this strategy because it is easier to hit a goal by aiming at target than by aiming at the goal directly. Specifically, by using intermediate and operating targets, it can more quickly judge whether its policies are on the right track, rather than waiting until it sees the final outcome of its policies on employment and the price level.

Suppose that the central bank's employment and price-level goals are consistent with a nominal GDP growth rate of 5 percent. If the central bank feels that the 5 percent nominal GDP growth rate will be achieved by a 4 percent growth rate for M2 (its intermediate target), which will in turn be achieved by a growth rate of $\frac{1}{2}$ percent for the monetary base (its operating target), it will carry out open market operations (its tool) to achieve the $\frac{1}{2}$ percent growth in the monetary base.

After implementing this policy, the central bank may find that the monetary base is growing too slowly, say, at a 2 percent rate; then it can correct this too slow growth by increasing the amount of its open market purchases. Somewhat later, the central bank will begin to see how its policy is affecting the growth rate of the money supply.

□ If M2 is growing too fast, say, at a 7 percent rate, the central bank may decide to reduce its open market purchases or make open market sales to reduce the M2 growth rate.

One way of thinking about this strategy (illustrated in Figure 5.1) is that the central bank is using its operating and intermediate targets to direct monetary policy toward the achievement of its goals. After the initial setting of the policy tools, an operating target such as the monetary base, which the central bank can control fairly directly, is used to reset the tools so that monetary policy is channeled toward achieving the intermediate target of a certain rate of money supply growth. Midcourse corrections in the policy tools can be made again when the central bank sees what is happening to its intermediate target, thus directing monetary policy so that it will achieve its goals of high employment and price stability (the spaceship reaches the moon)

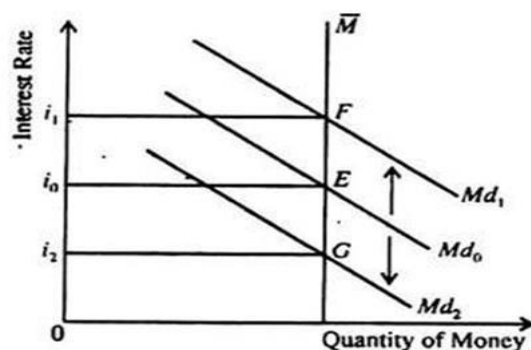


Figure 5.1 Result of Targeting on the Money Supply Targeting on the money supply at M^* , will lead to fluctuations in the interest rate between i' and i'' because of fluctuations in the money demand curve between M_d' and M_d'' .

iii. Choosing the Targets

There are two different types of target variables: interest rates and aggregates (monetary aggregates and reserve aggregates). In our example, the central bank chose a 4 percent growth rate for M2 to achieve a 5 percent rate of growth for nominal GDP. It could have chosen to lower the interest rate on the three-month Treasury bills to, say, 3 percent to achieve the same goal. Can the central bank choose to pursue both of these targets at the same time? The answer is no. Why a central bank must choose one or the other can be shown using the application of the supply and demand analysis of the money market.

Let's first see why a monetary aggregate target involves losing control of the interest rate. Figure 5.1 contains a supply and demand diagram for the money market. Although the central bank expects the demand curve for money to be at M_d^* , it fluctuates between M_d' and M_d'' because of unexpected increases or decreases in output or changes in the price level.

The money demand curve might also shift unexpectedly because the public's preferences about holding bonds versus money may change. If the central bank's monetary aggregate target of a 4 percent growth rate in M2 results in a money supply of M^* , it expects that the interest rate will be i^* . However, as the figure indicates, the fluctuations in the money demand curve between M_d' and M_d'' will result in an interest rate fluctuating between i' and i'' . Pursuing a monetary aggregate target implies that interest rates will fluctuate.

The supply and demand diagram in Figure 5.2 shows the consequences of an interest-rate target set at i^* . Again, the central bank expects the money demand curve to be at M_d^* , but it fluctuates between M_d' and M_d'' due to unexpected changes in output, the price level, or the public's preferences toward holding money.

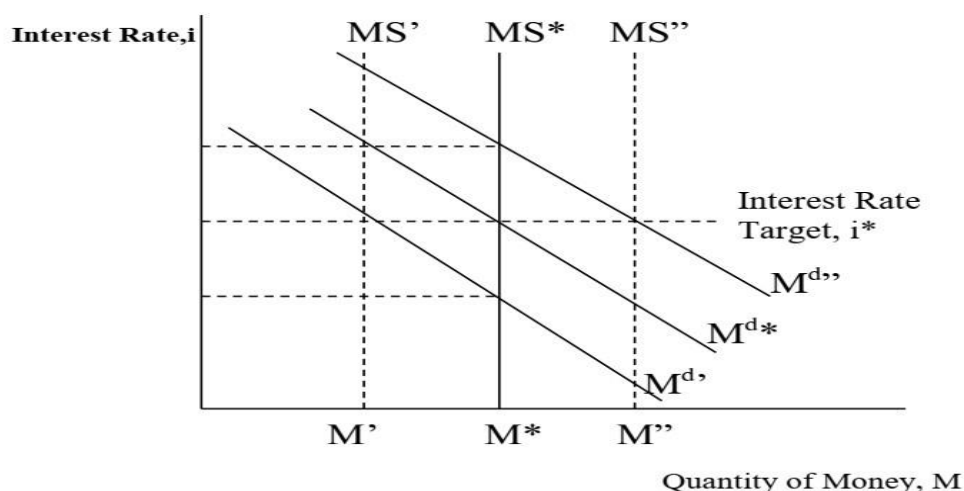


Figure 5.2 Result of Targeting on the Interest Rate

Targeting the interest rate at i^* will lead to fluctuations of the money supply between M' and M'' because of fluctuations in the money demand curve between M_d' and M_d'' .

If the demand curve falls to M_d , the interest rate will begin to fall below i^* , and the price of bonds will rise. With an interest rate target the central bank will prevent the interest rate from falling by selling bonds to drive their price back down and the interest rate back up to its former level. The central bank will make open market sales until the money supply declines to M_s' , at which point the equilibrium interest rate is again i^* . Conversely, if the demand curve rises to M_d'' and drives up the interest rate, the central bank would keep interest rates from rising by buying bonds to keep their prices from falling. The central bank will make open market purchases until the money supply rises to M_s'' and the equilibrium interest rate is i^* . The central bank's adherence to the interest-rate target thus leads to a fluctuating money supply as well as fluctuations in reserve aggregates such as the monetary base.

The conclusion from the supply and demand analysis is that interest-rate and monetary aggregate targets are incompatible: A central bank can hit one or the other but not both. Because a choice between them has to be made, we need to examine what criteria should be used to decide on the target variable.

iii. Criteria for Choosing Intermediate Targets

The rationale behind a central bank's strategy of using targets suggests three criteria for choosing an intermediate target: It must be measurable, it must be controllable by the central bank, and it must have a predictable effect on the goal.

Measurability: Quick and accurate measurement of an intermediate-target variable is necessary because the intermediate target will be useful only if it signals when policy is off track more rapidly than the goal. What good does it do for the central bank to plan to hit a 4 percent growth rate for M2 if it has no way of quickly and accurately measuring M2?

□ Data on the monetary aggregates are obtained after a two-week delay, and interest-rate data are available almost immediately. Data on a variable like GDP that serves as a goal, by contrast, are compiled quarterly and are made available with a month's delay. In addition, the GDP data are less accurate than data on the monetary aggregates or interest rates. On these grounds alone, focusing on interest rates and monetary aggregates as intermediate targets rather than on a goal like GDP can provide clearer signals about the status of the central bank's policy.

At first glance, interest rates seem to be more measurable than monetary aggregates and hence more useful as intermediate targets. Not only are the data on interest rates available more quickly than on monetary aggregates, but they are also measured more precisely and are rarely revised, in contrast to the monetary aggregates, which are subject to a fair amount of revision.

Controllability: A central bank must be able to exercise effective control over a variable if it is to function as a useful target. If the central bank cannot control an intermediate target, knowing that it is off track does little good because the central bank has no way of getting the target back on track. Some economists have suggested that nominal GDP should be used as an intermediate target, but since the central bank has little direct control over nominal GDP, it will not provide much guidance on how the Fed should set its policy tools. A central bank does, however, have a good deal of control over the monetary aggregates and interest rates.

Our discussion of the money supply process and the central bank's policy tools indicates that a central bank does have the ability to exercise a powerful effect on the money supply, although its control is not perfect. We have also seen that open market operations can be used to set interest rates by directly affecting the price of bonds.

Because a central bank can set interest rates directly whereas it cannot completely control the money supply, it might appear that interest rates dominate the monetary aggregates on the controllability criterion. However, a central bank cannot set real interest rates because it does not have control over expectations of inflation. So again, a clear-cut case cannot be made that interest rates are preferable to monetary aggregates as an intermediate target or vice versa. **Predictable Effect on Goals:** The most important characteristic a variable must have to be useful as an intermediate target is that it must have a predictable impact on a goal. □□ The central bank cannot use the price of tea in China to affect unemployment or the price level in its country. Because the ability to affect goals is so critical to the usefulness of an intermediate-target variable, the linkage of the money supply and interest rates with the goals output, employment, and the price level-is a matter of much debate.

iv. Criteria for Choosing Operating Targets

The choice of an operating target can be based on the same criteria used to evaluate intermediate targets. Both the federal funds rate and reserve aggregates are measured accurately and are available daily with almost no delay; both are easily controllable using the policy tools (such as open market operations, discount policy, reserve requirements).

When we look at the third criterion, however, we can think of the intermediate target as the goal for the operating target. An operating target that has a more predictable impact on the most desirable intermediate target is preferred. If the desired intermediate target is an interest rate, the preferred operating target will be an interest-rate variable like the federal funds rate because interest rates are closely tied to each other. However, if the desired intermediate target is a monetary aggregate, our money supply model in Chapter 4 shows that a reserve aggregate operating target such as the monetary base will be preferred. Because there does not seem to be much reason to choose an interest rate over a reserve aggregate on the basis of measurability or

controllability, the choice of which operating target is better rests on the choice of the intermediate, target (the goal of the operating target).

Chapter Six: Monetary Policy in Ethiopia

National Bank of Ethiopia which is the defacto sole regulator of the financial industry has adequate authority which is sufficient enough to accomplish its tasks and duties as a central bank. Moreover, it may seek power wherever appears necessary as the discretion is given to it by law. However, it is not well equipped enough to properly put into practice its powers and duties.

Ethiopian banking sector had not been directly affected by the international financial crisis due to its poor integration in the international financial markets, less stiff competition among the institutions, and its traditional banking system. However, this does not mean that Ethiopia had not been affected and will not be affected by the crisis. It does not also mean that the tomorrow's Ethiopian financial system will be the same with that of yesterdays. This calls for revisiting the adequacy of the existing supervisory capacity of National Bank of Ethiopia to forestall future banking crisis in Ethiopia.

6.1 Powers and Functions of the National Bank of Ethiopia

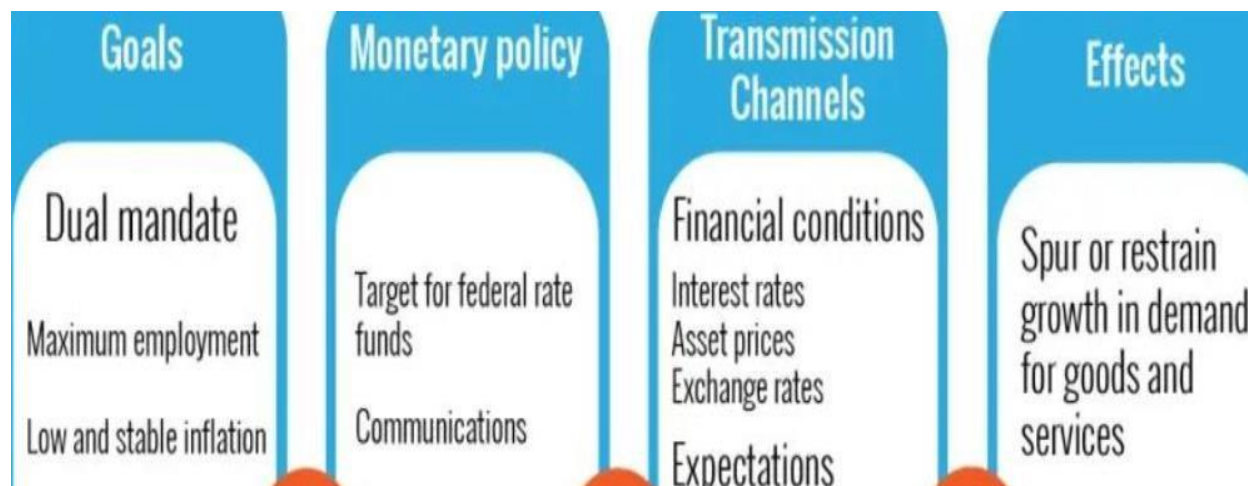
The Bank has the following powers and duties:

- Coins, prints and issues the legal tender currency, and regulates the country's money supply.
- Regulates the applicable interest rate and other cost of money charges.
- Formulating implements and follows-up the country's exchange rate policy, and manages and administers the international reserves of the country.
- Licenses, supervises and regulates the operations of banks, insurance companies and other financial institutions.

- Sets limits on gold and foreign exchange assets, which banks, and other financial institutions authorized to deal in foreign exchange uphold in deposits.
- Sets limits on the net foreign exchange positions and terms, and the amount of external indebtedness of banks and other financial institutions.
- Provides short and long term refinancing facilities to banks and other financial institutions.
- Accepts deposit of any kind from foreign sources.
- Promotes and encourages the dissemination of banking and insurance services throughout the country.
- Prepares periodic economic studies, together with forecasts of the balance of payments, money supply, prices and other relevant statistical indicators of the Ethiopian economy useful for analysis and for the formulation and determination by the Bank of monetary, saving and exchange policies.
- Acts as banker, fiscal agent and financial advisor to the Government.
- Represents the country in international monetary institutions and acts consistently with international monetary and banking agreements to which Ethiopia is a party.
- Exercises and performs such other powers and activities as central banks customarily perform.

6.2 Conduct of Monetary Policy in Ethiopia

- The principal objective of the monetary policy of the National Bank of Ethiopia is to maintain price & exchange rate stability and support sustainable economic growth of Ethiopia.
- Foster monetary, credit and financial conditions conducive to orderly,
- Balanced and sustained economic growth and development.
- The steps may involve the following



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